A Building Is Not Its Silhouette: Structural Blindness in 3D Generation Evaluation

Supplementary Material

A. Metrics and the Voxel Descriptor

We use two realism metrics throughout, both the unbiased kernel two-sample statistic of Bińkowski et al. [2] and differing only in the features they are computed on (Fig. 1).

- **MV-KID**, the *multi-view kernel distance*. Each structure is rendered from four orthographic views at 224px and embedded with DINOv2-B [21]. This is the render-and-embed family the field uses, and is our stand-in for it.
- **V-KID**, the *voxel kernel distance*. The same estimator on a 77-dimensional descriptor computed directly on the voxel grid, with no rendering and no GPU. The dimensions are four blocks: the distribution over the **55** D4-orbits of $2 \times 2 \times 2$ occupancy patterns (256 patterns fall into 55 orbits by Burnside's lemma under the order-8 group of yaw rotations and an x-mirror), a **5**-bin wall-thickness histogram, an **8**-bin height profile normalized by the build's own height, and **9** global scalars. The group is exactly D4: a vertical flip is deliberately *not* quotiented out, since upside-down is a real defect in a building, and we confirm it moves 45 of the 77 dimensions. Patterns are counted over the volume padded by one air cell, so surface patterns are counted rather than clipped at the border. The all-empty orbit is identically zero after normalization, leaving 76 informative dimensions and, since each of the three histograms sums to one, 73 free parameters.

Estimator and kernel. Both metrics are the unbiased estimator of MMD^2 with the cubic polynomial kernel of KID, $k(a, b) = (a^\top b/D + 1)^3$, where D is the feature dimension (768 for MV-KID, 77 for V-KID). Diagonal terms are excluded from both within-set sums, and values are reported multiplied by 10^3. Every feature row is unit norm before it reaches the kernel, so $\gamma = 1/D$ sits in the same range for both metrics. The MMD-RBF row of the ladder uses a Gaussian kernel with its bandwidth set by the median heuristic.

Render and camera specification. Each build is cropped to its bounding box and meshed as one textured quad per block face bordering air. The textures are the vanilla Minecraft client's block textures, sampled nearest-neighbour, with flat unlit shading and Minecraft-style face brightness baked into the tiles (1.0 top, 0.85 north and south, 0.70 east and west, 0.55 bottom). The four cameras are orthographic, at azimuths 45°, 135°, 225° and 315° and elevation 30°, each framed to the projected bounding box with a 5% margin, rendered at 224×224 (pyrender, EGL) and composited onto white. DINOv2-B (`facebook/dinov2-base`) receives each view through the checkpoint's own image processor; we take the CLS token of the last layer, L2-normalize it per view, average the four views and normalize again, so the sample unit is the build. A render canary, four fixed builds hashed at cache time, invalidates the feature cache whenever the rendered pixels change.

The nine global scalars. Mean and 90th-percentile taxicab depth of occupied voxels below the surface; surface-to-volume, exposed faces per block; wall fraction, the share of exposed faces lying in a coplanar connected patch of at least 9 faces; yaw symmetry, the better mirror IoU over the two horizontal axes; the sealed and the aperture-aware interior ratios, interior air over bounding-box volume (Sec. B); height entropy, the entropy of the 8-bin height profile; and $\log(1+n_{\text{blocks}})$. The thickness histogram bins taxicab depth $1, 2, 3, 4, \geq 5$.

Whitening and clipping. Descriptor blocks differ by orders of magnitude in scale, so the feature map whitens against statistics frozen on the reference, **clips at ± 5 reference standard deviations**, and then L2-normalizes each row. The clip is load-bearing rather than cosmetic and we report it for that reason: real houses have almost no mass five voxels deep, so the reference standard deviation of the thickness ≥ 5 bin is 1.6×10^{-4}, and FILL-INTERIOR moves that single bin by 67.5 reference standard deviations before clipping. Without the clip one coordinate would dominate the kernel, and the separation we report would be a one-dimensional thickness test wearing a 77-dimensional coat.

Both metrics are reported in *noise floors*: the spread of repeated real-vs-reference draws at the same n. A value of 1 means one sampling-noise unit away from real, which makes the two commensurable despite differing by three orders of magnitude in raw units. Fig. 2 walks through the descriptor.

B. Probes and Counting Interiors

Sealed air. A cell is *sealed* air when it is empty and not 6-connected through empty cells to the boundary of the build's bounding box. FILL-INTERIOR fills exactly these cells with the build's most common material. HOLLOW-SHELL keeps a block only if one of its six faces

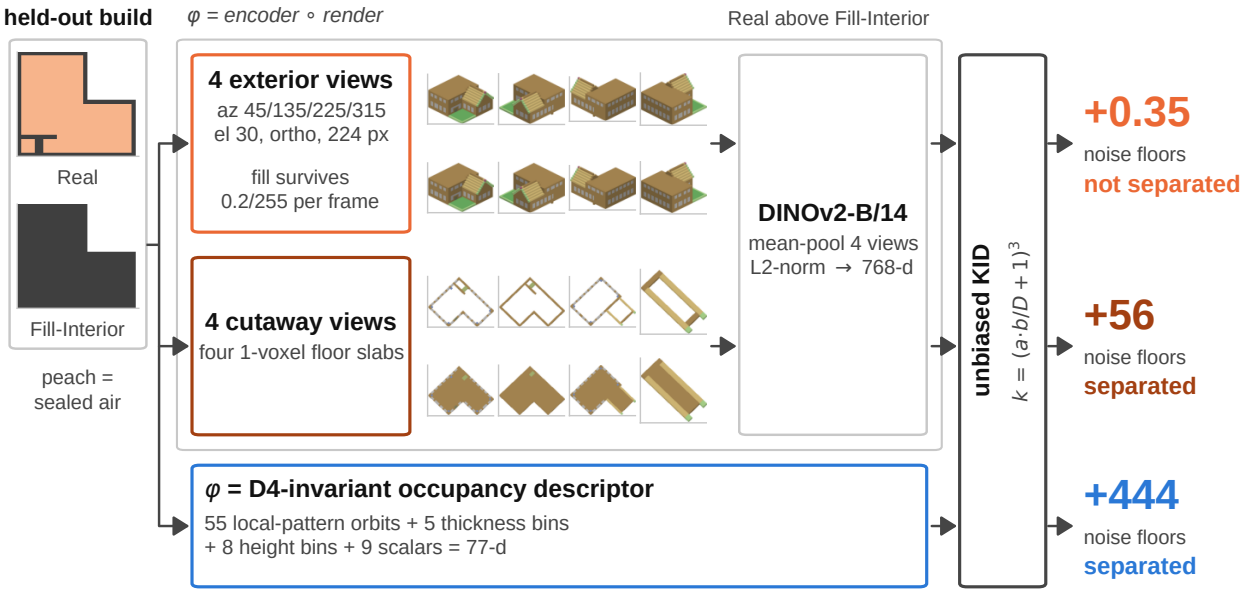


Figure 1. **One estimator, three feature maps.** MV-KID and V-KID are the same unbiased kernel distance of main paper Eq. 1 and differ only in ϕ. *Row 1:* the render-and-embed path, where the structure reaches the statistic as four images. *Row 2:* the same renderer and the same encoder with cutaway views, which recovers the signal. One renderer, one encoder, only the views differ, and four other encoders and 8 or 16 views change nothing (Table 10). *Row 3:* the geometric path, which reads occupancy directly and needs no renderer and no GPU. A noise floor is a run's own real-vs-real spread and falls with n, so separations compare within a run: rows 1 and 3 come from the n=256 leaderboard against 399 reference builds, and row 2 from the view ablation at n=256 against 512, where row 1 scores +0.2. The pixel residue in row 1 is the mean over the 128 exterior frames of the converse-probe run, not one build. The sections at left are horizontal slices of one 3D-Craft build.

borders air 6-connected to the outside, so it removes the solid behind the exterior shell, where FILL-INTERIOR fills the sealed air behind it.

Counting rooms behind a doorway. A naive enclosed-air count, restricted to sealed air, counts a room with an open doorway as no room at all. We additionally close the solid under a 26-connected morphological closing that seals openings up to two blocks wide and report the air that becomes enclosed only then. Calibrated on a synthetic hollow cube with a 343-cell room: a 1×1, 1×2 or 2×2 doorway recovers all 343 cells; a 3×3 hole or a missing wall recovers 0, correctly classifying an open structure as having no room. On the same 200 held-out houses the probe statistics of main paper §4.1 are measured on, **6.0**% have no *sealed* interior at all, and *every one* of those twelve builds does have a room reachable through a doorway. Per build, the sealed share of a build's own interior volume has a median of **100**% and a mean of **85.5**%: most houses are fully sealed, and the rest lose a whole room to an open door. Pooled over builds, apertures account for **25.6**% of total real interior volume, which is the margin by which a sealed-only counter understates real houses. Every claim in the earlier literature that generated builds "have no interiors" was made with that counter.

The dose axis. PARTIAL-FILL fills interior cells *inward from the walls*, nearest-first, which gives an exactly calibrated dose (interior ratio 0.1175 → 0.0941 → 0.0708 → 0.0475 → 0.0242). Random speckle and largest-room-first filling were rejected, since the first reads as noise and the second removes 71% of the interior at a nominal 25% dose. Fig. 3 shows both probe directions on three builds.

C. Validation Gates

Every candidate metric is scored on rungs of known damage in units of its own noise floor, and a metric that fails a gate is withheld from ranking (main paper §4.3). This section gives the render tier's ladder, the procedural baselines, the provenance of each gate and the defects the gates themselves had.

C.1. The render ladder

Table 1 applies the gates to the render tier, Table 2 gives the seven render-space statistics on the LEGO ladder, and Fig. 4 places both metrics on eight rungs of one run. The per-metric ladder tables express a rung as its distance from the null *mean*, while the leaderboard expresses an arm as

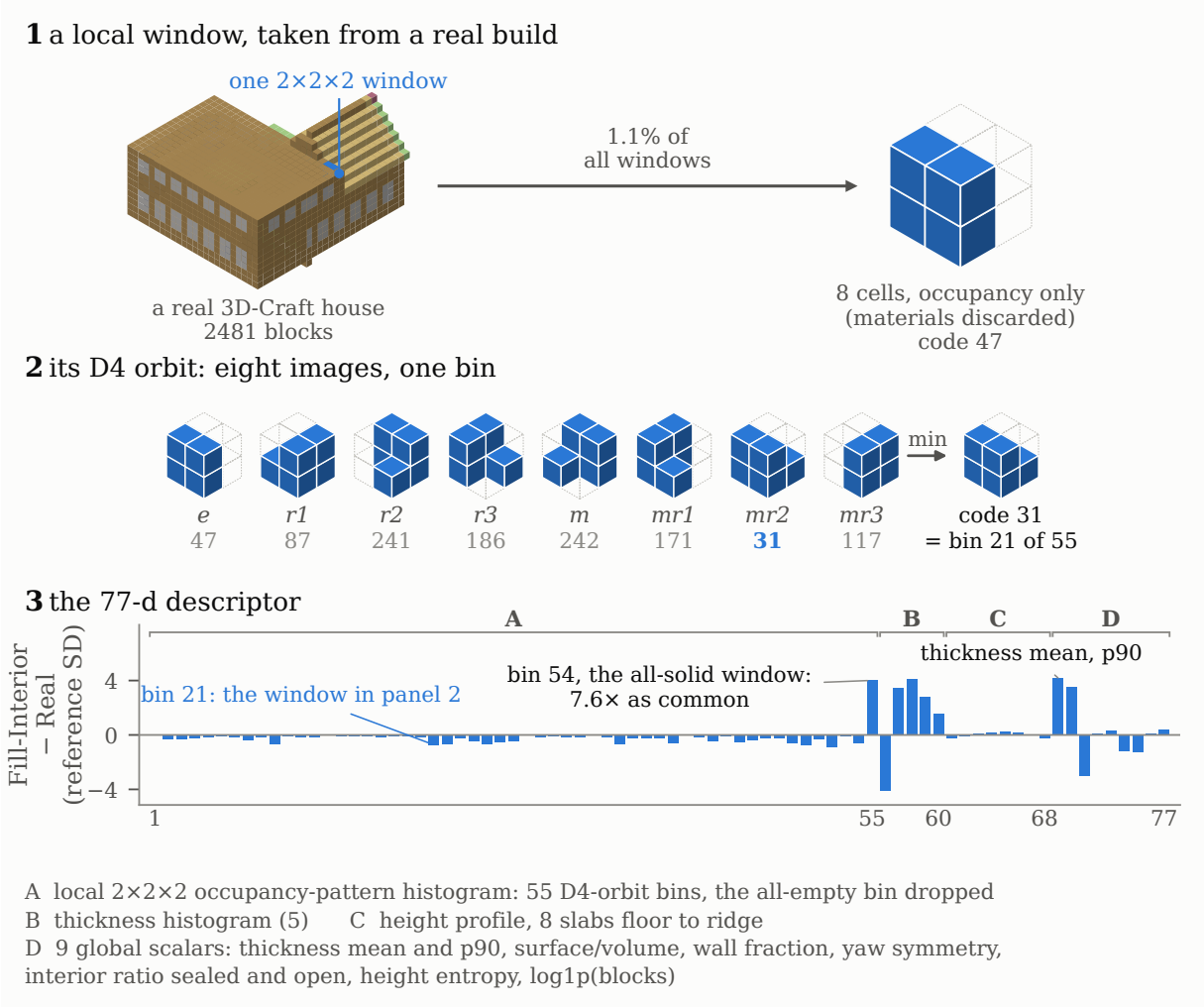


Figure 2. **The geometric descriptor.** A $2 \times 2 \times 2$ occupancy neighborhood, its orbit under the order-8 group of yaw rotations and an x-mirror collapsing to one canonical bin, and the resulting feature vector with the shift FILL-INTERIOR induces. The thickness histogram loses its one-block walls and gains thick cores, and surface-to-volume with both interior ratios falls. Filling every sealed interior moves this descriptor 202 noise floors at n=128, where the same builds move the render tier 0.14.

its distance from a held-out REAL *draw*. On an exactly invariant rung the two conventions differ by construction: the geometry metric's material-shuffle row is 0 exactly against the REAL draw and 1.19 against the null mean, so the conventions must not be mixed within one comparison.

C.2. Procedural baselines

Why coherence is reported as distance to real. Two rows at the extremes of main paper Tab. 4 show why the suite reports more than its two realism pillars. An arm of *verbatim copies of the training set* scores 0.012 raw MV-KID, third of sixteen arms, behind only held-out real data and FILL-INTERIOR, and only the novelty columns catch it: nearest-neighbor IoU 1.000, duplicate rate 1.000 and a retrieval percentile of 0.000 against a real held-out floor of 0.507. At the other end of the scale the procedural baselines are *too* hollow, with aperture-aware interior ratios of 0.61 (HOLLOW-BOX) and 0.56 (GABLED-HOUSE) against a real 0.13. A coherence measure read as "higher is better" would therefore rank a hollow box above a house, which is why every coherence measure is reported as distance to real.

C.3. Where the gates come from, and whether they favour V-KID

V-KID and the gates that admit it come from one project, so the gates need evidence that they were not shaped around it. We give three kinds: each gate's prior use, when each threshold was set, and the verdicts of Table 13 under alternative thresholds.

Table 1. The render tier's metric-validation ladder (DINOv2-B, 4 orthographic views at 224px, n=64). Columns are the probe rungs: real held-out, retype 10% of blocks (the rung the ordering gate uses for this tier), displace 10%, delete 40%, material shuffle and a 90° rotation; **sd** is the metric's own noise floor. Gates are blocking, and *only **KID** and **MMD-RBF** pass*: PRDC coverage fails n-stability, and recall fails noise ordering and invariance. MMD-RBF is on its own scale and is reported to one decimal; the last row is the biased estimator. The *real* column scores a held-out draw against the reference, a different null from the two disjoint halves of one real draw in main paper §4.3, which is why the unbiased KID does not read 0 here.

metric	real	retype 10%	displace 10%	del. 40%	mat. shuffle	rot. 90°	noise floor	gate
KID (unbiased)	**0.011**	**0.165**	**0.435**	**0.106**	**0.762**	**0.010**	**0.008**	**PASS**
MMD-RBF	**3.5**	**49.6**	**137.4**	**32.2**	**218.0**	**3.0**	**2.4**	**PASS**
1−recall	0.137	0.331	0.899	0.371	0.790	0.105	0.028	FAIL: noise ord., inv.
Fréchet distance	0.107	0.157	0.250	0.147	0.373	0.106	0.005	FAIL: n-stab.
1−coverage	0.246	0.508	0.835	0.488	0.948	0.242	0.057	FAIL: n-stab.
1−density	0.000	0.247	0.637	0.166	0.897	0.000	0.073	FAIL: noise ord.
1−precision	0.125	0.234	0.234	0.109	0.641	0.125	0.036	FAIL: noise ord.
biased MMD estimator	0.079	0.468	1.148	0.317	1.987	0.075	0.021	FAIL: n-stab.

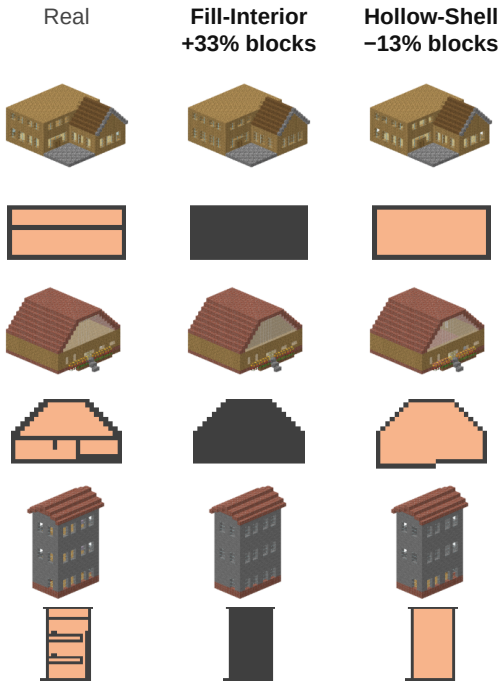


Figure 3. **Both directions of the corruption, on three held-out builds.** FILL-INTERIOR adds a median 33% blocks and HOLLOW-SHELL removes a median 13%, yet the views the metric is given are near-identical in every column. The cross-sections show what changed, with sealed interior air in peach. In the converse-probe run at n=256, MV-KID places both within one noise floor of real (+0.21 and −0.02); V-KID places them 402 and 231 away, against 444.36 and 265.4 on the leaderboard's own floor.

Table 2. The render-tier ladder on the LEGO domain (StableText2Brick, DINOv2-B, 4 views, n=128), each statistic in its own noise floors, measured from the floor's mean rather than from a held-out draw, so these values are not directly comparable with Table 10's or with main paper Tab. 3, where the same probe reads −0.09 on MV-KID at n=253. **All seven miss FILL-INTERIOR and resolve every other corruption.**

statistic	FILL-INTERIOR	20% deleted	MATERIAL-SHUFFLE
KID	**2.0**	12.3	132.3
MMD-RBF	**2.1**	13.9	148.8
Fréchet distance	**0.5**	10.8	72.4
1−coverage	**0.2**	7.7	19.8
1−density	**0.1**	7.2	14.0
1−precision	**−0.4**	9.0	28.0
1−recall	**0.4**	1.4	30.7

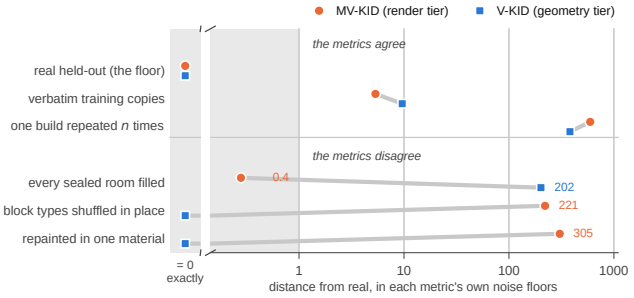


Figure 4. **Two metrics, one run, eight rungs of known damage.** Each rung's distance from held-out real, in that metric's own noise floors (n=128, symlog). The geometric metric's two material rungs are bit-identical to real, not merely small, so they sit in a separate gutter. The band marks one noise floor; per-statistic numbers are in Table 2.

Prior use. Table 4 maps each gate to earlier work that states the same criterion. Ordering under graded corruption is the test FID was introduced with [9], repeated for KID [2] and made a rank-correlation criterion for graph metrics [20, 28]. Invariance to quality-preserving transforms is a stated desideratum [3, 4], and FID's violations of it

Table 3. Procedural baselines, with real held-out data for reference. GABLED-HOUSE is the bar that matters: what a competent afternoon of hand-written rules produces. PATCH-STITCH is an attack, not a baseline. Every cell is a per-build mean over n=128 builds per baseline; the real row is all 399 held-out test builds. **Interior is reported twice**, because the two counters answer different questions (Sec. B): *sealed* is air unreachable from the bounding box, *open* additionally credits rooms reachable only through a door or window. The distinction is not cosmetic: GABLED-HOUSE has a doorway, so it scores 0.000 sealed and 0.562 open.

				interior		
baseline	what it is	blocks	wall frac	sealed	open	LCC
RANDOM-VOXELS	random voxels at real density	1383	0.049	0.002	0.003	0.406
HOLLOW-BOX	hollow box, floor, walls, flat roof	1345	1.000	0.609	0.609	1.000
PATCH-STITCH	8^3 patches from *different* builds	1405	0.624	0.003	0.009	0.866
GABLED-HOUSE	box, pitched roof, door, windows	1139	0.969	0.000	0.562	1.000
TRAIN-COPY+NOISE	a training build, 10% displaced	1057	0.454	0.002	0.055	0.967
real	held-out test split	924	0.745	0.114	0.129	0.978

are documented [14, 22]. A metric that cannot recognise two samples of one distribution fails the test Naeem et al. [19] apply to precision and recall, and sample-size bias is the defect Bińkowski et al. [2], Chong and Forsyth [6] and Jayasumana et al. [11] identify in FID. The memorization gates follow the held-out-referenced data-copying test of Meehan et al. [17], and the within-set twin gate follows the approximate-copy diversity test of Jiralerspong et al. [12]. Text-metric checklists apply the same directional and invariance tests [24, 25]. What we change is the use: a failing metric is withheld from ranking rather than reported with a caveat. O'Bray et al. [20] use graded perturbations to select metric hyperparameters, never to disqualify a metric, and without an invariance or sample-size test; Borji [3] list the desiderata without running them as a gate. The one prior meta-evaluation of metrics in this domain, Hervé and Salge [8], correlates eighteen procedural-content metrics with GDMC jury scores, the route a human-validated version of this suite would take.

gate	prior statement of the criterion
G1, G2 ordering	Bińkowski et al. [2], Heusel et al. [9], Jayasumana et al. [11], Jiralerspong et al. [12], O'Bray et al. [20], Thompson et al. [28], Xu et al. [29]
G6 invariance	Borji [3, 4], Kynkäänniemi et al. [14], Parmar et al. [22], Ribeiro et al. [24]
G7 self-consistency	Bińkowski et al. [2], Lopez-Paz and Oquab [15], Naeem et al. [19], Yang et al. [30]
G8 n-stability	Bińkowski et al. [2], Chong and Forsyth [6], Jayasumana et al. [11], Jiralerspong et al. [12], Thompson et al. [28]
memorization	Alaa et al. [1], Jiralerspong et al. [12], Meehan et al. [17], Somepalli et al. [26], Stein et al. [27]
diversity, twins	Jiralerspong et al. [12], Naeem et al. [19], Xu et al. [29]

Table 4. **Each gate restates a published criterion.**

When the thresholds were set. The ordering, invariance, self-consistency and n-stability thresholds were committed and validated on render metrics four weeks before the first voxel-descriptor metric existed, and the gate code, with the 77-dimensional descriptor, has been unchanged for 58 days before the head-to-head of Sec. I.2 ran. The only later change to the gates is the pair of denominator corrections of Sec. C.4, which apply to every metric; the twin gate reuses the duplicate gate's rule.

Verdicts under other thresholds. We recompute every gate of Table 13 from the stored scores and, for V-KID, occupancy KID and the six IoU rows, from a rerun that reproduces the table to 2×10^{-6} and also records each null mean and drift. Each threshold moves over a range around its value: ordering tolerance from -1 (each step must rise by one floor) to $+2$ (a step may fall by two), the invariance bar from 0.5 to 5 floors, and the self-consistency and n-stability bars from 1 to 5.

- At every ordering tolerance ≤ 0 and every invariance bar, with the other two bars at their values, V-KID is the only row that passes (Fig. 5).
- V-KID passes in all 360 of the 480 joint settings with a self-consistency bar of at least 2 floors. A bar of 1 floor rejects every row, V-KID included, as a single draw lies within one floor of its mean only about two thirds of the time.
- Its margins are wide: the smallest ordering step is 7.5 floors, invariance holds exactly, the self-consistency deviation is 1.2 floors and the drift is 0.4 standard errors.
- The rows a looser threshold admits are those whose failure is a tie rather than an inversion. Allowing a step to fall by half a floor admits 1-NNA on IoU and EMD, which

separate FILL-INTERIOR by 2.7 and 1.6 floors; adding an invariance bar of 2 floors admits occupancy KID (48.4). No threshold admits an encoder as a geometry metric, since MATERIAL-SHUFFLE and MONOCHROME move them by 63 to 106 floors, nor IoU nearest-neighbour MMD, whose n-drift is 34 to 36 standard errors.

The same holds for disqualification. Across novelty bars of 2, 3 and 5 spreads and floors of half, one and two times their values, both positive controls, Dream-Cubed and the hollow-box baseline are disqualified in all nine settings. Only the loosest corner readmits an arm, the gabled-house baseline, and tightening to 2 spreads first disqualifies our own autoregressive arms on DINO nearest-neighbour percentile.

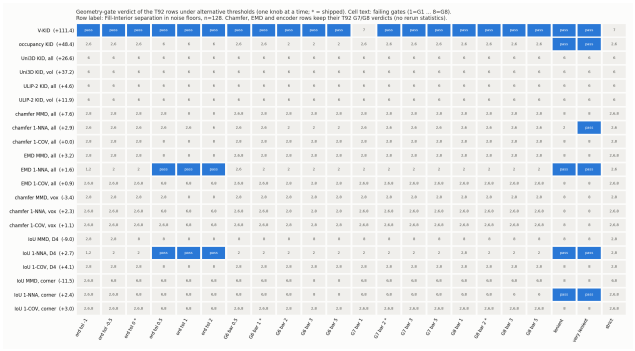


Figure 5. **Gate verdicts under alternative thresholds.** Rows of Table 13 with their FILL-INTERIOR separation; columns move one threshold at a time ($*$ marks the value used) or all four together. Blue: passes every gate on the geometry rungs. Text: the gates that fail.

C.4. Auditing the gates

A gate needs auditing like any other measurement, and ours had two denominator defects and one fail-open renderer. The noise floor, the unit every gate is stated in, was first estimated from subsamples of a pool that excluded the probe set and carried no finite-population factor; corrected, a metric that separates FILL-INTERIOR at 67 s.d. passes the self-consistency gate it had failed. The n-stability gate first tested a difference of means against the spread of a single draw, larger by $\sqrt{\text{reps}}$; against the standard error of the difference, the biased estimator's $1/n$ decay is rejected at 14.8 standard errors. Separately, the renderer once substituted a blank frame for any rendering exception, so a library mismatch made every arm score KID $= -4\times10^{-13}$, better than real. It now raises past a 2% failure rate, and the repaired library reproduced the stored render canary bit-for-bit, so no recorded number is affected.

D. Aggregation and Statistical Reading

D.1. What BlockScore's maximum is

BlockScore is the maximum of five pillar scores, and a maximum of several mean-zero quantities is not itself mean-zero. The control arm reads exactly 0.00 on every pillar because it *is* the arm the calibration is taken from: $(v_p - v_p)/\mathrm{sd}_p = 0$ identically, with zero variance. That is an identity, not a measurement, and main paper Tab. 4 labels it as such. A *fresh* real arm, drawn from the same distribution but not used to calibrate, scores $\mathbb{E}[\text{BlockScore}] = 1.08$ with standard deviation 0.89, right-skewed (+0.89), and a 95th percentile of 2.8. Read BlockScore against that null rather than against zero: a value of 15 is not "15 standard deviations" but "far outside a null whose upper tail is under 3". The inflation is about one noise floor, against a smallest score of 229.29 on the leaderboard, so it changes no ordering and no conclusion in this paper. A future suite comparing arms that sit within a few floors of real would need to subtract it.

The five pillars are not in one unit. Appearance, geometry and palette are divided by a genuine sampling noise floor, and we verify this: their declared units match the spread of fresh real draws to within 7% (0.93, 1.03 and 1.05 of the measured value). The coherence and geometry-scalar pillars are divided by a per-build robust spread of the reference, an effect size measuring how far apart two distributions are in units of how much individual builds differ. Measurement puts that unit 29× and 52× coarser than those pillars' own sampling noise. A structure pillar would therefore have to move some 4,800 of its actual noise floors to veto a live pillar sitting at +166, and on our card none ever does. We state this as a limitation of the current calibration rather than a property of the aggregation rule: the maximum is the right rule, and two of the five inputs are quoted in a unit that makes them diagnostics rather than vetoes.

Because the winning pillar's unit is what the number is in, BlockScore is uninterpretable without the pillar that produced it, which is why every leaderboard we print carries a *worst pillar* column. It is also not convertible to a p-value: the null is max-shaped and skewed, so the usual normal reading does not apply.

D.2. The noise-floor unit depends on n

Separations throughout this paper are quoted in *noise floors*: the standard deviation of repeated real-versus-reference draws at the same sample size. That unit makes two metrics with incommensurable raw scales comparable within a run, which is why we use it. It is not scale-free across runs, and we state the dependence explicitly because it is easy to misread. Measured on the identical corruption at two sample sizes in the dose run (399 reference builds, its own 48-draw floor):

metric	sd n=64	sd n=128	ratio	sep n=64	sep n=128	ratio
V-KID	0.1732	0.0848	**0.49**	132.9	286.7	**2.16**
MV-KID	0.00438	0.00316	0.72	0.48	0.34	0.71

The noise floor of the geometric metric halves when n doubles, the $1/n$ decay expected of a degenerate U-statistic under the null, while the population effect is fixed. A separation in noise floors therefore grows *linearly* in n, and a figure such as "202" means "202 at n=128". The n=256 leaderboard measures the same probe against the same reference at 444.36, a factor of 2.2 (Sec. D.3). Separations are therefore only comparable across runs at equal n, which is why our protocol pins it, and regenerating every arm at a larger n leaves the *ordering* of arms intact while roughly doubling every magnitude. MV-KID scales less cleanly (0.72 rather than 0.49) because its separation is a ratio of two quantities that are both near zero, and the render row should be read as "indistinguishable", not as a number.

The unit is itself estimated. The floor comes from 48 real-versus-reference draws and carries a relative standard error near $1/\sqrt{2 \cdot 47} \approx 10\%$, so a separation should be read to one or two significant figures with its run named. We quote 202 because it is the protocol run's value; a sweep over twelve seeds places it between 185 and 274. The dose run above draws its own floor against the same reference (0.0848 at n=128 against the protocol run's 0.110) and reads the same probe at 286.7, and the validation ladder, which scores against a 248-build subset and measures from the null mean, reads 111.4 (Table 13).

The linear-in-n rule holds at fixed reference size. The doubling above, 202 at n=128 to 444 at n=256, assumes the reference pool is unchanged. Our view-protocol ablation measures the same probe on the same corpus and split at n=256 and reads 120.8, not 400, because it scores against 512 reference builds rather than 399 and its floor (0.188) is consequently *larger* than the n=128 floor rather than half of it. The rule describes how a separation moves when n grows within one protocol; it does not survive a change of reference, which is why every number in this paper is reported with its run.

D.3. The n=256 leaderboard and the n=128 protocol run

The leaderboard of main paper Tab. 4 reruns the protocol run's board at the protocol's own n=256, with the same reference, split, 77-dimensional descriptor and seed. The generator arms were extended from 128 to 256 samples from the same checkpoints and sampling stream, so the first 128 samples of every arm are byte-identical to the protocol run's.

Table 5. BlockScore and rank at n=128 (protocol run) and n=256 (leaderboard). Raw kernel distances barely move; the noise floor roughly halves, so every BlockScore roughly doubles. Ranks are the leaderboard's own; ties as in main paper Tab. 4. The n=256 BlockScore of an arm the within-set near-twin gate disqualifies (DQ) is the maximum pillar it would have scored (Sec. L.1); the gate postdates the protocol run, so the n=128 column is ungated.

arm	BlockScore		rank	
	n=128	n=256	128	256
Dream-Cubed	—	165.71	—	DQ
VOXEL-AR (contaminated)	106.87	229.29	2	2
VOXEL-AR (clean)	136.73	285.22	3	3
VOXEL-AR-U (clean)	DQ	345.10	DQ	4
MATERIAL-SHUFFLE	231.52	422.43	7	5–7
PATCH-STITCH	246.15	431.70	8	5–7
FILL-INTERIOR	201.80	444.36	6	5–7
MONOCHROME	297.95	535.36	9	8
TRAIN-COPY+NOISE	302.73	619.13	11	9
GABLED-HOUSE	300.15	620.98	10	DQ
HOLLOW-BOX	476.22	864.00	12	DQ
RANDOM-VOXELS	603.09	1087.06	13	10

Table 5 sets the two boards side by side. The noise floor roughly halves at n=256 (0.110 → 0.053 for V-KID, 0.00289 → 0.00161 for MV-KID) while the raw distances hold (VOXEL-AR clean: V-KID 14.90 → 15.06, MV-KID 0.245 → 0.254), so separations double and only ranks and ratios carry across. Three changes go beyond the doubling. VOXEL-AR-U's memorization gate was marginal at n=128, +3.0 spreads against a bar of 3.0; at n=256 it reads +2.1 and the arm ranks fourth. The within-set near-twin gate disqualifies Dream-Cubed, GABLED-HOUSE and HOLLOW-BOX (Sec. L.1). Among the controls, MATERIAL-SHUFFLE, PATCH-STITCH and FILL-INTERIOR become a three-way tie (adjacent flip probabilities 0.20 and 0.21). The rank intervals are 95% intervals over 1,000 replicates that subsample 80% of every arm and share one reference subsample per replicate; a pair is a tie when its order flips in at least 5% of replicates.

Calibrating the finite-population correction. The floor at n=256 draws 256 of the 399 test builds per replicate and divides the spread of those overlapping draws by $\sqrt{1 - n/N}$. For a degenerate U-statistic the leading variance term carries $(1 - f)^2$ rather than $(1 - f)$, which would make this correction too small, but the pool's own kernel matrix adds a first-order term of order ζ_2/N that more than offsets it. We measure the net effect against non-overlapping draws: a reference of 399 and a population of 1,464 real builds, both drawn from the training split so that the statistic sits in the same near-degenerate regime, with the true spread read off disjoint n-sets of the population.

The true spread is 0.88 to 0.95 of the bench floor at n=256 and 0.89 to 0.93 at n=128, for both metrics, so the floor is slightly conservative; a $(1-f)$ correction would overstate it by a factor near 1.8. On the leaderboard's own pool an unbiased Hoeffding estimate, accurate to 5% in the simulation, puts the stored V-KID floor 11% above the true one and the stored MV-KID floor 2.5% below it. The larger term is the Monte-Carlo error of Sec. D.2: 48 draws pin the V-KID floor at n=256 only to $[0.043, 0.067]$ at 95%, which is why two runs scoring the same raw distance against two 48-draw floors can read MATERIAL-SHUFFLE at +422 and at +346. We keep each run's stored floor; under the corrected floor no gate verdict of V-KID or MV-KID changes and no ranking changes.

D.4. The head-to-head test

Arm pairs are compared with an exact label-permutation test: pool the two arms, re-split at the observed sizes, and recompute the difference of kernel distances against the same fixed reference, with Holm correction [10] over the family of pairs. A resampling estimate of the standard error does not work here, because the difference is a degenerate U-statistic when both arms match the reference and a non-degenerate one when they do not, so no single rescale is correct. The permutation null tracks a Monte-Carlo ground truth within 5% in both regimes and its type-I error is nominal at both levels. An earlier subsampling test rejected 3.4× too often on matched nulls and 7.2× on mismatched ones, and a floored bootstrap p-value earlier made every pair unseparable after Holm; both are replaced.

Recomputing the run of record changes only claims of separation, which is the expected direction: a test that rejects too easily can manufacture a difference but never a tie. Both of our null results hold and strengthen: FILL-INTERIOR remains unseparated from held-out real (p from 0.61 to 0.77), and the clean retrain remains unseparated from the language-model baseline (0.89 to 0.93). Two positive separations do not survive: TRAIN-COPY against held-out real falls to the α=0.05 boundary, and the contaminated arm separates from one language-model baseline but not the other. We report the blindness result by magnitude for this reason, and the case study's conclusion, that the contaminated arm's apparent advantage does not survive a clean split, is unaffected, since it rests on a within-architecture comparison.

Non-detection, and equivalence. Every probe is built from the same builds as the real arm it is compared with. The label-permutation test treats the two arms as independent, which is exact under its null but discards the pairing, so on a probe it is conservative: FILL-INTERIOR against real reads p=0.87 at n=256. A paired test that swaps labels within each build and its probe reads p=0.055 for the same +0.35 floors, and 0.82 for HOLLOW-SHELL. Across n from 8 to 256 the paired test never reaches 80% power on either interior probe under MV-KID (at most 0.20), while V-KID detects both at n=8 (Fig. 6). Non-rejection is not equivalence, so we also test equivalence against a margin fixed in advance, two noise floors, with two one-sided tests whose standard error is that of a fresh draw of 256 builds and of the reference, from the Hoeffding decomposition of the paired difference. FILL-INTERIOR reads +0.35 floors with 90% interval $[-0.28, +0.98]$ and HOLLOW-SHELL +0.04 with $[-0.25, +0.33]$, both equivalent to real (p<0.001); the same holds for a margin of 0.05 in effect size (±2.7 floors). Under V-KID neither is equivalent, at 444 and 265 floors.

D.5. Reading a squared statistic over a population of prompts

How a metric's value is *read* is a separate issue from what it can see. Because KID and MMD are squared, an arm that fails on a quarter of its builds scores only 6.0% of the way to fully bad, where a reader assuming proportionality would infer a quarter of the damage. For a squared-MMD statistic the response to a mixture is quadratic,

$$\mathrm{MMD}^2(\alpha P + (1-\alpha)Q,\ P) = (1-\alpha)^2\,\mathrm{MMD}^2(Q, P), \tag{1}$$

a one-line consequence of the linearity of the mean embedding. Bińkowski et al. [2] state this identity in their Appendix B.2, and both they and Heusel et al. [9] plot their metrics against a contamination fraction. We add the reporting recommendation it implies: report $\mathrm{MMD} = \sqrt{\mathrm{MMD}^2}$, which is *linear* in the contaminated fraction and free to compute. The identity holds for squared integral probability metrics such as KID and MMD. It does not hold for Fréchet distance, which Bińkowski et al. [2] show is not polynomial in α, so we derive the correction for KID only.

Measured on a mixture ladder whose bad pool is RANDOM-VOXELS at n=128, the best-fit exponent of MV-KID is **2.05** against the identity's 2. An arm failing on a quarter of its builds scores 6.0% of the way to fully bad (0.105 against 1.740). Reporting $\sqrt{\mathrm{KID}}/\sqrt{\mathrm{KID}_{\mathrm{worst}}}$ recovers the true bad fraction to a mean absolute error of **0.013** (MV-KID) and **0.020** (V-KID), against 0.128 and 0.131 for the raw values.

One caution on the readout. The identity holds for the population quantity, while we report an unbiased estimator of MMD^2, which can be negative and whose square root is therefore biased downward by Jensen's inequality, worst where the statistic is small relative to its own standard error. We report the square root as a diagnostic for reading a failure rate, never as the ranked quantity.

An effect size for comparing metrics. A separation in noise floors is a test statistic: it grows linearly in n

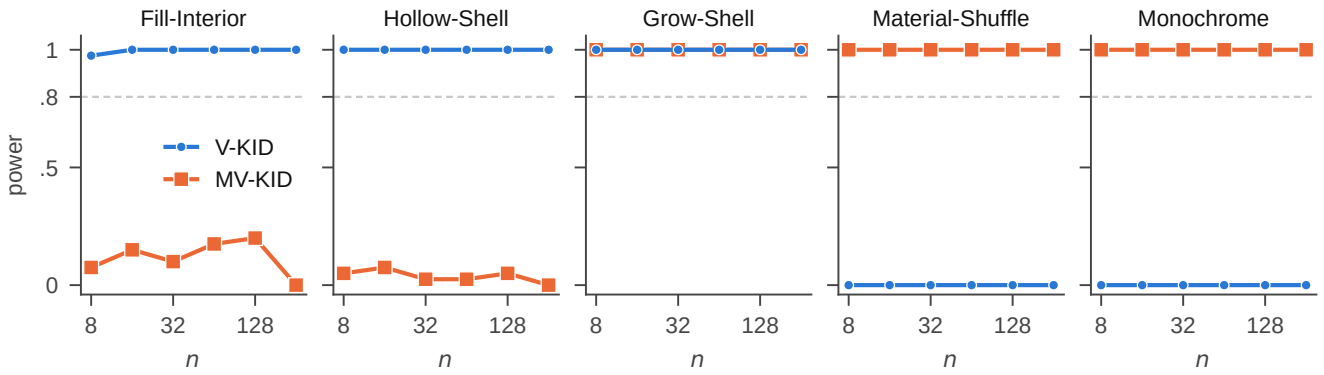


Figure 6. **Detection as a function of n.** Power of the paired permutation test (α=0.05, 999 permutations) of each probe against its own source builds, over 40 random subsets per n below 256 and the full set at n=256; dashed line, 80%. V-KID detects FILL-INTERIOR, HOLLOW-SHELL and GROW-SHELL at n=8. MV-KID detects GROW-SHELL, MATERIAL-SHUFFLE and MONOCHROME at n=8 and never reaches 80% power on FILL-INTERIOR or HOLLOW-SHELL.

(Sec. D.2) and at a different rate per metric, so a ratio of two metrics' separations is not a ratio of what they see. The readout above gives a quantity that does not grow with n, $e = \sqrt{\mathrm{MMD}^2_{\mathrm{arm}}} / \sqrt{\mathrm{MMD}^2_{\mathrm{RANDOM\text{-}VOXELS}}}$ under the same metric, which reads 0 for real data and 1 for the anchor whatever the metric. Table 6 gives it for every probe at n=256. The two interior probes sit at 0.03 and 0.02 of the anchor on appearance, inside the 95th percentile of fresh real draws (0.031), and at 0.78 and 0.60 on geometry. MATERIAL-SHUFFLE and MONOCHROME are the converse. PATCH-STITCH sits as far from real on geometry as on appearance, so its near-real local statistics buy it no geometric advantage. Read between a probe and its own source builds, the render features of FILL-INTERIOR do move, by 0.018 $[0.014, 0.024]$ of the anchor against a paired null of 0.005: that is the glass leak of Sec. H, 44 times smaller than V-KID's 0.80 on the same pairs.

Table 6. **Effect size** e, $\sqrt{\mathrm{MMD}^2}$ over that of RANDOM-VOXELS under the same metric, n=256 against the 399-build reference, with half-sampling 95% intervals. The 95th percentile of fresh real draws is 0.031 (MV-KID) and 0 (V-KID). BrickGPT is scored on the LEGO suite, where the anchor is SINGLE-MODE.

arm	MV-KID	V-KID	ratio
FILL-INTERIOR	0.03 [0.00, 0.08]	0.78 [0.74, 0.83]	27
HOLLOW-SHELL	0.02 [0.00, 0.08]	0.60 [0.57, 0.63]	26
GROW-SHELL	0.47 [0.44, 0.50]	0.80 [0.77, 0.84]	1.7
MATERIAL-SHUFFLE	0.62 [0.60, 0.65]	0	0
MONOCHROME	0.70 [0.68, 0.73]	0	0
PATCH-STITCH	0.63 [0.61, 0.66]	0.71 [0.69, 0.73]	1.1
BrickGPT	0.17	0.37	2.2

E. Corpus, Split and Arms

E.1. Corpus construction

Three public sources are pooled onto a shared legacy block vocabulary (the text2mc dump is remapped with 97% block-state coverage; world cuts are ground-stripped), passed through a quality gate, and deduplicated variant-aware into `houses_32`, the corpus every reported run uses (main paper §5).

Table 7. Corpus construction, transcribed from the cache manifest. Per-source columns are post-gate, post-deduplication counts and sum to **final**, which is *pooled* less quality drops less exact duplicates. Material variants, the same shape in different materials, are *kept*.

pooled	drops	dups	**final**	GrabCraft	3D-Craft	text2mc
3,493	826	6	**2,661**	1,360	1,267	34

Drop reasons, 826 in total: no interior 675, single-material blob 52, fragmented 37, too flat 29, fewer than three materials 22, too sparse 9, too few blocks 2.

The enclosed-air gate is calibrated per source. The gate requires at least 8 interior air voxels unreachable from the bounding box boundary. Measured on ground truth, 17% of GrabCraft houses legitimately have open interiors, against 31% of 3D-Craft and 54% of text2mc, so GrabCraft is exempted from the gate and the other sources are not. What the gate removes from the gated sources is trees, a truck and roof fragments, read off a pre-gate visual grid.

E.2. Arms

Table 8 gives every arm with its recipe.

VOXEL-AR. VOXEL-AR is a 5.66M-parameter decoder-only transformer (6 layers, width 256, 8 heads, feed-forward width 1,024, dropout 0.1) with learned token, block-index and phase embeddings. It reads a build as a sequence of 3D-BPE pieces: a vocabulary of 1,079 merged pieces learned on the training split, keeping stair, log and door facings as distinct tokens, at a maximum length of 5,480 tokens. It is trained on the 1,863-build canonical training split under orientation-aware D4 augmentation, 13,890 sequences of median length 1,814, for 60 epochs at batch 8 with AdamW at learning rate 3×10^{-4} and bf16 autocast, about two hours on one GPU, and sampled at temperature 1.0 with top-k 40. VOXEL-AR-U drops the facing tokens and the orientation-aware augmentation. The contaminated checkpoint has the same architecture and recipe and was trained on an earlier split (Sec. L.2).

What is not a control. Real data requantized to a coarser grid is not a ranked control and not a ladder rung: decimation changes a structure's resolution, not its quality, so it is not a severity axis, and no result in this paper rests on it.

E.3. The calibration draw

The real-vs-real draw that defines the noise floor must be a shuffled sample of the test split. Taking the first n test builds instead inflates the measured noise floor by $\mathbf{10.9}\times$ (s.d. 0.0320 against 0.0029 at n=128) and shifts its center from +0.0001 to +0.0410. A contiguous block of a corpus stored in source and family order is not an exchangeable sample of it, so the floor it produces is both wider and displaced, and every separation measured against it is understated. Controls are re-scored in every run, and the protocol pins corpus, split, n, reference size, control size and seed; runs that do not match are listed with the reason and not ranked.

F. Dose–Response

Table 9 gives every rung of the dose-response sweep and Fig. 7 plots the three axes.

G. View Protocols

G.1. Encoders, view budgets and cutaways

Table 10 varies the encoder, the number and projection of the views, and the cut, holding the statistic fixed.

G.2. Facade and cross-section views

The corner views of MV-KID are one choice of exterior observation. We test two others, holding the builds, the reference and the encoder of the n=256 leaderboard fixed (every arm reproduces the leaderboard's build hashes). The V-KID column uses the leaderboard's own noise floor. Each view condition draws its own floor from 48 fresh real draws, and for the corner views that draw comes out 1.22 times the leaderboard's, so from identical raw distances the corner column reads MATERIAL-SHUFFLE at +346 where the leaderboard reads +422.43; on the leaderboard's floor the corner column reads FILL-INTERIOR at +0.35, HOLLOW-SHELL at +0.04 and MONOCHROME at +535.36. *Facade views* are the views an architect draws: four side elevations at zero elevation, alone (A-sides) and with a roof plan (A-render), and the 2.5D facade representation itself, five images of the first block along each axis shaded by its depth, drawn without a renderer (A-2.5D). *Cross-section views* open the build: one-voxel sections through the middle of x, z and y seen face-on (B-mid), sections at 25, 50 and 75% of each axis (B-9), and the build with half removed along each axis, seen at the cut (B-cut). Table 11 reports every condition in its own noise floors above the held-out real draw.

Facade views do not see the interior. The 2.5D facade images of FILL-INTERIOR and HOLLOW-SHELL are bit-identical to those of the real builds for all 256 houses, so the separation is exactly zero. The representation records the first block along each axis and counts glass as a block, and a probe that keeps the exterior air region fixed keeps every first block fixed. This is Lemma 2 with no renderer involved. Rendered elevations read FILL-INTERIOR at +4.4 noise floors, above the corner views' +0.3, because a face-on camera looks straight through windows into the filled rooms; at n=128 the same condition reads +2.3. The signal is a window leak, a hundred times smaller than the 444 of V-KID, and HOLLOW-SHELL stays below one noise floor.

Cross-sections see the interior through appearance. A one-voxel section shows rooms as background and filled rooms as material, and B-mid separates FILL-INTERIOR by +71 and HOLLOW-SHELL by +40 noise floors. Removing half of the build (B-cut) leaves the far walls in view and reaches +14. Both are far below V-KID, and the ordering of the other arms barely moves: every view condition ranks the eleven arms as the corner views do ($\rho \geq 0.93$) and unlike V-KID ($\rho \leq 0.28$), with MONOCHROME and the procedural baselines furthest from real. Showing the inside to an image encoder recovers the interior signal without turning the metric into a structural one, which is the complementary blindness that motivates scoring each pillar separately (main paper §6.4).

Invariance. Axis-aligned views are not invariant to the heading of a build unless the pooled set is. Under a 90° yaw and a mirror of the held-out real builds, every condition moves by at most 0.5 noise floors, so mean-pooling over the sides is invariant in practice.

Table 8. Every arm, with its full recipe. The compact version is main paper Tab. 1 in main paper §5.

kind	name	what it is
floor	REAL	held-out real builds from the test split, disjoint from the reference they are scored against
probe	FILL-INTERIOR	every enclosed air cell filled with the build's own dominant material
probe	MATERIAL-SHUFFLE	the block multiset permuted across the build's own occupied cells: geometry exactly preserved
probe	MONOCHROME	every block replaced by one material: palette destroyed, geometry exactly preserved
probe	TRAIN-COPY	verbatim copies of training builds
probe	SINGLE-MODE	one build repeated n times
baseline	RANDOM-VOXELS	random voxels at the real occupancy density
baseline	HOLLOW-BOX	a hollow box: floor, walls, flat roof
baseline	GABLED-HOUSE	box, pitched roof, door, windows: a competent afternoon of rules
baseline	PATCH-STITCH	8^3 patches lifted verbatim from *different* real builds
baseline	TRAIN-COPY+NOISE	a training build with 10% of its blocks displaced
submission	VOXEL-AR	autoregressive transformer over merged-piece tokens, trained on the canonical split
submission	VOXEL-AR-U	the same model without orientation augmentation
submission	Dream-Cubed	a published masked discrete diffusion model; its village checkpoint's largest constructed component per chunk (Sec. L.1)
case study	LLM-FT	a fine-tuned code language model emitting one line per piece; protocol run only
case study	LLM-FT-GEO	LLM-FT with a learned relative-3D attention bias; protocol run only

Table 9. Dose-response, n=64 and n=128 per rung. **range (sd)** is the best-to-worst span in units of the metric's own noise floor, given as n=64 / n=128. The render metric is perfectly monotone on two axes and *flat* on the third; on that axis its rank correlation (+0.700 at both n) is the correlation of a sequence that does not move, so read the span instead. The mixture axis replaces real builds with RANDOM-VOXELS.

axis	metric	Spearman	monotone	range (sd)
solidify (interiors filled, 0 → 100%)	V-KID	+1.000	yes	**133 / 287**
	MV-KID	+0.700	**no**	**0.48 / 0.34**
occupancy noise (0 → 20% blocks moved)	V-KID	+1.000	yes	226 / 455
	MV-KID	+1.000	yes	143 / 185
mixture (fraction of arm that is real)	V-KID	+1.000	yes	211 / 444
	MV-KID	+1.000	yes	392 / 549

H. What a Render Protocol Provably Cannot See

A deterministic render pipeline R_V with a fixed view set V induces an equivalence relation on structures: $x \sim x'$ exactly when $R_V(x) = R_V(x')$.

Lemma 1 (Render kernel)**.** *If $x \sim x'$ then $T(\mathcal{E}(R_V(x))) = T(\mathcal{E}(R_V(x')))$ for every encoder $\mathcal{E}$, every pooling rule and every statistic T.*

Proof. Immediate: the composition is a function of $R_V(x)$ alone. □

On its own Lemma 1 is a tautology. It becomes a statement about this paper because FILL-INTERIOR lands inside an equivalence class, which needs two further facts.

Lemma 2 (Surface invariance)**.** FILL-INTERIOR *leaves the exterior-air region unchanged, and leaves every exterior face unchanged in cell, direction and material.*

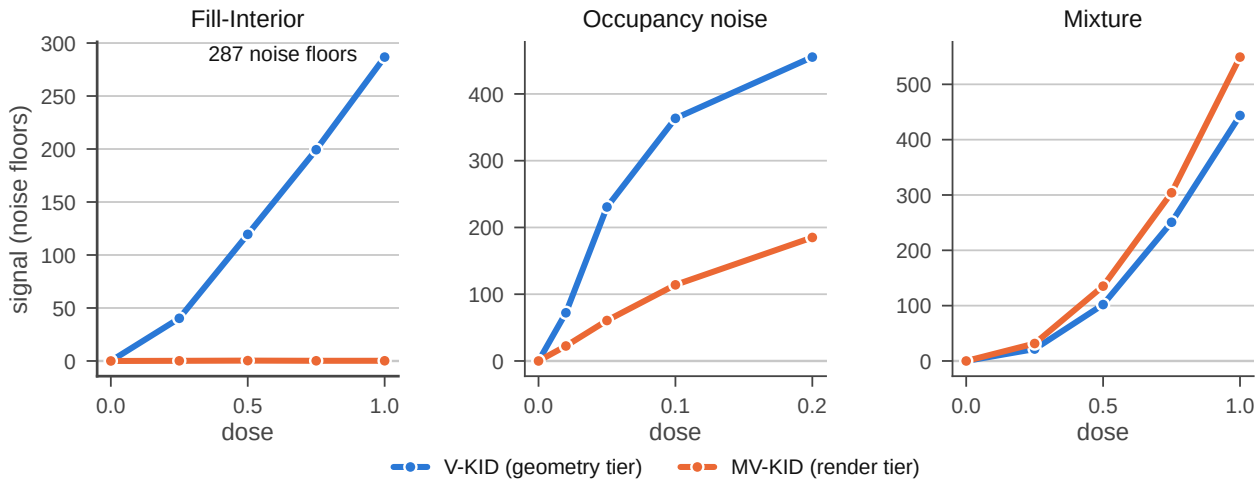


Figure 7. **Dose–response.** Each metric in units of its own noise floor, which is the only scale on which the two are commensurable. Both rise monotonically on occupancy noise and on the mixture axis; on the interior axis the render metric does not move.

Table 10. Separation of FILL-INTERIOR from real in each condition's own noise floors (n=256 against 512 reference builds, paired subsample, 95% intervals). **Every exterior condition lands between 0.0 and 0.2 noise floors; cutaways reach 7–56 with the same encoder and the same statistic.**

	condition	separation	interior?
encoder	DINOv2-B (ours)	+0.2 [0.1, 0.3]	no
	CLIP ViT-B/16	+0.1 [−0.1, 0.2]	no
	CLIP ViT-B/32	+0.1 [0.0, 0.2]	no
	ResNet-50 (ImageNet-supervised)	+0.0 [−0.1, 0.1]	no
	Inception-v3 pool3 (the FID backbone)	+0.2 [0.0, 0.3]	no
views	8 orthographic	+0.2 [0.1, 0.3]	no
	16 orthographic	+0.1 [−0.0, 0.2]	no
	perspective	+0.1 [−0.0, 0.2]	no
cut	**facing quadrant removed**	**+6.7** [5.9, 7.5]	**yes**
	vertical cross-sections	**+14.0** [12.8, 15.2]	**yes**
	roof removed	**+17.1** [15.3, 19.2]	**yes**
	floorplan cross-sections	**+55.8** [52.4, 59.6]	**yes**
	V-KID (no rendering at all)	+120.8 [115.7, 126.3]	yes

Lemma 3 (Occlusion)**.** *Assume (a) nearest-surface shading, (b) faces bordering exterior air are opaque, and (c) every camera lies outside the bounding box. Then* $R_V(\text{FILL-INTERIOR}(x)) = R_V(x)$ *for* every *view set* V.

Proof sketch. Any ray reaching a newly filled cell must first cross a face bordering exterior air. By (b) that face is opaque and by (a) it is what the ray returns, so no pixel can depend on the filled interior. □

Lemma 2 has a consequence beyond rendering. Any metric computed on samples of the *exterior surface* is a function of that face set, so it too is constant under FILL-INTERIOR. We verify it: sampling exterior face centers from a build and from its solidified copy gives **identical point sets on 128 of 128 builds**, so chamfer distance, earth-mover distance and every other point-cloud distance between them is exactly zero. This matters for the proposal to replace renders with surface samples [16], and for this corpus in particular: the model released with it [5] reports a cumulative chamfer distance. Sampling *all* exposed faces, including those bordering sealed cavities, does see the probe (mean chamfer 0.485), but a surface sampler that reached interior cavity walls would be observing the interior, which is the thing at issue.

	corner	A-sides	A-render	A-2.5D	B-mid	B-9	B-cut	V-KID
FILL-INTERIOR	+0.3	+4.4	+3.6	0	+71.1	+67.5	+14.3	+444.4
HOLLOW-SHELL	+0.0	+0.8	+0.6	0	+39.7	+42.8	+5.3	+265.4
MATERIAL-SHUFFLE	+346	+548	+610	+355	+226	+292	+415	0
MONOCHROME	+439	+1146	+1309	+314	+1080	+1314	+925	0
yaw 90°	−0.2	+0.0	+0.1	+0.1	+0.4	+0.5	−0.2	0
mirror	−0.2	+0.1	+0.1	+0.0	−0.4	−0.3	+0.0	0
ρ vs corner	1	0.97	0.97	0.96	0.93	0.93	0.95	0.27
ρ vs V-KID	0.27	0.10	0.10	0.28	0.07	0.07	0.05	1

Table 11. **Facade and cross-section views** (n=256)**.** Separation in noise floors above held-out real data. ρ: Spearman correlation of the ordering of eleven arms (four probes, three procedural baselines, three autoregressive arms, Dream-Cubed) with corner-view MV-KID and with V-KID. The A-2.5D images of FILL-INTERIOR and HOLLOW-SHELL are bit-identical to the real images for all 256 builds.

Together with Lemma 1 this is an impossibility result, not a statistical one: under an opaque renderer *no* encoder, statistic, sample size or view budget separates a corpus from its solidified copy. We verify Lemma 2 exhaustively (0 violations over the 399 held-out builds, and 0 for HOLLOW-SHELL) and Lemma 3 by rendering all 1,596 test-split views through counterfactual pipelines that switch each leak off in turn. *Fixed atlas* packs one texture atlas for

the whole experiment, so no texture coordinate depends on which faces a build has:

renderer	bit-identical	px changed (med.)	max ch. (med.)	MV-KID, n=256
glass opaque, fixed atlas, CPU raster (Lemma 3)	**100.0%**	—	—	**0**
glass opaque, fixed atlas, GPU raster	88.9%	0.008%	1	−0.00004
glass opaque, per-build atlas	8.1%	0.99%	79	—
translucent, fixed atlas	41.3%	0.21%	107	+0.40
deployed (translucent, per-build atlas)	9.1%	1.23%	128	+0.29

Our renderer violates hypothesis (b) in two ways, and the two have different standing. The first is physical: Minecraft glass is translucent, so a window in front of a filled void reveals the fill. This is a real channel, and it alone alters 59% of views. The second is an artifact of our mesh builder, which we report because it inflates that count. `build_mesh` assigns texture-atlas tiles in face-iteration order over *all* exposed faces, interior ones included; deleting interior faces can change the tile order, hence the atlas layout, hence every UV rectangle, and the sampler is nearest-neighbor. With glass opaque this alone alters 92% of views. A reader should therefore take 91% as a property of our renderer, not of the corruption; the corruption's own reach is the 59% that survives fixing the atlas. Counted per house, the fill shows in at least one view of 67% of houses with the atlas fixed and of 92% under the deployed renderer. The GPU rasterizer adds a third, numerical channel: it is not bit-reproducible, and rendering the same build twice changes up to 9 pixels by $1/255$ in 6 of 24 houses. Across the test split this reaches at most $2/255$ on 0.01% of pixels and moves MV-KID by 4×10^{-5} noise floors; the CPU rasterizer has none. On LEGO, whose wool is opaque, the deployed renderer's −0.01 for FILL-INTERIOR becomes −0.00004 with the atlas fixed.

We then rescored the probe end to end through each of the four pipelines, holding the arms, the draw and the estimator fixed and changing only the texture pack. This turns the lemma into a prediction the benchmark either satisfies or does not:

renderer	MV-KID separation	features identical
opaque, flat tiles	**+0.000**	**yes**
flat tiles, translucency on	−0.069	no
textured, translucency off	+0.026	no
deployed	+0.119	no

The prediction holds exactly: bit-identical renders give bit-identical features and therefore an identical kernel distance, so the separation is 0 as an identity rather than as a rounded measurement. The other three rows carry the more useful lesson. The residual the benchmark reports for this probe is not a small signal. It is *wobble*. Changing only the texture pack, which changes nothing whatever about the geometry being probed, moves it from +0.119 to +0.026 to −0.069: a range wider than the value itself, and one that crosses zero, with the probe reading *closer* to real than real does. This is what it looks like when a metric is flat on an axis rather than weakly sensitive to it, and it is why we argue the point by magnitude and ordering rather than by the residual's sign. (The deployed row reads +0.119 here against the protocol run's +0.14 for the same raw values, −0.0018 and −0.0014; the two runs estimate their own floors, 0.00337 and 0.00289, which is the effect Sec. D.2 documents.)

Where a leaked signal is lost is a measurement rather than a theorem. In DINOv2 the patch tokens a changed pixel touches move by 12% of the between-build scale, the CLS token by 3.9% and the four-view pooled embedding by 2.0%, so the signal enters the features and is attenuated by aggregation rather than ignored by the encoder.

The converse holds by the same style of argument and makes the paper's complementarity claim provable on both sides rather than measured on one.

Lemma 4 (Material blindness of V-KID)**.** *The descriptor of Sec. A factors through the occupancy mask, so any transformation preserving occupancy,* MATERIAL-SHUFFLE *and* MONOCHROME *among them, leaves it pointwise unchanged, and V-KID returns the real arm's value exactly.*

Verified: descriptors are bit-identical (max difference 0.0) and V-KID returns the identical float for all three arms. The +0.00 printed in main paper Tab. 4 is an identity, not a rounded measurement. Renders of those same builds are never identical (0 of 256 views; a median 25% of pixels move), so material permutation provably escapes the render kernel exactly as interior fill provably falls inside it.

I. Other Observers: Point Clouds, Structural Metrics, a VLM Judge

I.1. 3D-native metrics and their samplers

We score two learned point-cloud encoders, Uni3D-B and ULIP-2 PointBERT (10k points with color, each repository's preprocessing), and two point-set distances, chamfer on 2,048 points and exact EMD on 512, each under three samplers. *all* samples uniformly by area over every face, the default of Uni3D, ULIP-2, PointFlow-style MMD and FPD. *vis* samples only faces an exterior orthographic ray hits first, which is what a watertight-mesh or rendered-depth pipeline observes. *cut* samples the faces the four cutaway views show. Minecraft builds become meshes with one quad per block face bordering air. The Objaverse set is 290 buildings passing a strict multi-room rule (hidden area ≥ 30%, hidden volume ≥ 5%, at least two sealed spaces), scored at n=96 against 130 reference objects; *strip* removes every hidden face and *fill* replaces the hidden volume by a solid, with an exterior pixel change of $0.04/255$. The Minecraft ladder is scored at n=128 against 248 reference builds. Cells are

null-standardized z-scores, seed 0 / seed 1; seed changes the split, the probe draw and every sampler draw.

The reading is the same on both domains. Under *vis* the learned encoders still separate exterior damage at 5 to 188 floors, so they are not insensitive; they are given no interior to see. Under *all* and *cut*, Uni3D sees filled Minecraft interiors at 22 to 37 floors and ULIP-2 more weakly. Chamfer and EMD stay at or below about 8 floors even when their points include the interior. Generated meshes from surface-fitted or score-distillation generators are exterior shells, so on them *all* and *vis* coincide and a learned 3D metric is blind in practice as well. Seed agreement is Pearson 0.91 over 275 Objaverse cells and 0.98 over 336 Minecraft cells.

I.2. V-KID against structural alternatives

Render metrics are blind to interiors by construction, so outscoring them is not what a geometry pillar must show. We therefore compare V-KID against the structural metrics a 3D reviewer would reach for, under the tests V-KID itself passed. The rows are:

- kernel distances on the 3D encoders Uni3D and ULIP-2, fed points from every exposed block face, sealed rooms included (*all*), or uniform points inside occupied voxels (*vol*);
- the set statistics of the shape-generation literature, MMD, 1−coverage and 1-NN accuracy, under chamfer and exact EMD on all-face points, chamfer on voxel centres, and 1−IoU between occupancy grids, either aligned at the bounding-box corner or maximised over the eight yaw and mirror variants;
- KID on the occupancy grid itself, bounding-box centred in 32^3 and 4^3 average-pooled. This is the same estimator as V-KID, with raw occupancy as the features.

Each row is scored by the ladder code that gates the leaderboard. That ladder uses the houses_32 ladder subset, n=128, 48 noise draws and the geometry rungs. The encoders, which read colour, are also judged on the render rungs. Each row is then scored on the n=256 arms of the leaderboard of record, rebuilt with matching build hashes. The V-KID row reproduces the leaderboard exactly (+444.4).

Of these rows, only V-KID passes every gate. It separates FILL-INTERIOR 2.3× more strongly than the next row and HOLLOW-SHELL 5× more strongly, and appearance and rotation leave it unmoved. The nearest alternative is occupancy KID: the same estimator on the raw grid. It sees FILL-INTERIOR and orders the leaderboard much as V-KID does (ρ=0.83), but it fails noise ordering. Its rotation reading also sits at the edge of the invariance bar, because a centred grid is not yaw-invariant. What V-KID adds is the D4-invariant local descriptor, not the estimator, and generic orbit-averaged occupancy descriptors pass as well (Sec. J), so the class that passes is the D4-invariant occupancy kernel distance, of which V-KID is one instance.

3D encoders see the interior once their points include it, but read appearance. On interior-inclusive points Uni3D and ULIP-2 separate both probes, so their blindness in Table 12 belongs to the visible-surface sampler and not to the encoders. They are not geometry metrics, however. MATERIAL-SHUFFLE and MONOCHROME, which leave occupancy unchanged, move them by up to 106 noise floors at n=128. Uni3D is not yaw-invariant (a 90° turn moves it on volume points by +5.7 floors at n=256), and both rank the leaderboard as MV-KID does ($\rho \geq 0.78$). As a geometry pillar they would let a generator trade structure for appearance, which the worst-pillar aggregation exists to prevent.

The voxel-generation suite misses the probes or rewards the fill. Chamfer, EMD and IoU set statistics never separate the interior probes by more than single digits, and each fails noise ordering or n-stability. Nearest-neighbour MMD under IoU or voxel-centre chamfer goes further and places FILL-INTERIOR *closer* to the reference than held-out real data, by 3 to 12 noise floors. A filled house overlaps its neighbours more than a house with rooms does. The metric the voxel literature reports would therefore reward a generator for filling its interiors.

Cost. On about 3,000 to 4,400 builds, V-KID takes 15 to 18 s on one CPU process. Each encoder takes 25 to 50 s of GPU time after sampling, chamfer takes 2 to 3 minutes of GPU time, and exact EMD on 512 points took 32 minutes on 16 CPU cores.

I.3. The VLM judge

The judge is `gpt-4o-2024-11-20` at temperature 0, shown two builds as four exterior (or cutaway) views each and asked which is the more realistic house, with both presentation orders and a plan registered before any call. Only 3D-Craft builds and probes derived from them reach the API. 2,288 trials, all valid. Each build is one image, its four views at azimuths 45°, 135°, 225° and 315° in a 2×2 grid. The system prompt is "You are an experienced Minecraft player judging builds for a research benchmark. You always answer with a single JSON object." The user message is the question, then "Build A:" and its image, then "Build B:" and its image, where the question reads "Two Minecraft builds, A and B, are shown below. <view note> Which looks more like a real Minecraft house, as built by a player? You must pick one; "equal" is not allowed. Reply with only a JSON object: {"choice": "A" or "B", "reason": "<at most 20 words>"}". The view note is "Each image shows one build from four viewing angles (a 2x2

Table 12. **Every 3D-native metric inherits the null space of its sampler.** Separation from held-out real (noise floors, seed 0 / seed 1). Under *vis* every point-cloud row is blind to both interior probes on both domains, within 1.6 floors in every cell; under *all* and *cut* the learned encoders see them.

Objaverse multi-room buildings, n=96 metric	strip	fill	noise	missing wall	decimate
DINOv2, 4 ext. views, KID	+0.0 / +0.0	+0.0 / +0.0	+34.7 / +66.6	+9.5 / +15.0	+21.3 / +29.0
DINOv2, 4 cutaways, KID	+10.9 / +9.0	+13.5 / +13.7	+43.9 / +32.7	+7.0 / +6.2	+22.3 / +14.6
V-KID	+23.7 / +26.5	+73.8 / +118.4	+18.1 / +31.8	+16.2 / +13.4	+12.0 / +12.4
Uni3D KID, *all*	+4.6 / +11.5	+8.5 / +14.1	+13.7 / +34.0	+9.8 / +15.4	+4.3 / +6.7
Uni3D KID, *vis*	+0.4 / −0.2	+0.3 / +0.0	+9.5 / +20.7	+16.5 / +18.5	+7.9 / +8.3
Uni3D KID, *cut*	+4.3 / +8.4	+6.3 / +12.4	+3.7 / +13.2	+5.6 / +10.0	+1.9 / +2.9
ULIP-2 KID, *all*	+2.4 / +4.2	+3.3 / +8.4	+5.0 / +19.6	+12.5 / +29.7	+7.8 / +14.0
ULIP-2 KID, *vis*	−0.0 / −0.3	+0.1 / −0.1	+6.5 / +16.3	+15.2 / +22.4	+9.3 / +10.9
ULIP-2 KID, *cut*	+4.2 / +4.7	+4.9 / +10.8	+3.5 / +11.5	+8.4 / +18.8	+5.1 / +6.0
chamfer MMD, *all*	+3.4 / +3.9	−0.6 / −0.4	−0.3 / +0.0	+9.9 / +10.9	+0.2 / +1.4
chamfer MMD, *vis*	−0.1 / +0.2	−0.0 / +0.1	−0.6 / −0.6	+7.9 / +10.5	−0.9 / +0.3
chamfer MMD, *cut*	+2.8 / +2.9	−0.7 / −0.1	−0.2 / +0.1	+8.8 / +7.7	+0.2 / +0.9
EMD MMD, *all*	+3.4 / +3.8	+0.7 / +0.2	+3.7 / +2.3	+19.6 / +20.5	+0.9 / +1.7
EMD MMD, *vis*	−0.0 / +0.2	−0.2 / +0.3	+0.5 / +0.9	+11.6 / +9.9	+0.4 / +0.6
EMD MMD, *cut*	+2.2 / +3.3	−0.1 / −0.1	+0.6 / +0.8	+13.2 / +15.2	+0.5 / +1.0
Minecraft houses, n=128 metric	FILL-INTERIOR	HOLLOW-SHELL	canon-16	40% deleted	MATERIAL-SHUFFLE
Uni3D KID, *all*	+21.7 / +22.6	+8.5 / +9.6	+154.8 / +186.0	+50.9 / +65.4	+16.5 / +18.9
Uni3D KID, *vis*	−0.1 / +0.4	+0.1 / +0.4	+128.2 / +117.1	+46.0 / +43.1	+19.0 / +16.0
Uni3D KID, *cut*	+37.0 / +30.4	+13.5 / +10.4	+132.3 / +117.9	+43.6 / +49.8	+19.1 / +15.8
ULIP-2 KID, *all*	+4.0 / +3.2	+2.1 / +1.9	+70.2 / +63.0	+49.1 / +48.1	+33.6 / +27.0
ULIP-2 KID, *vis*	−0.1 / +0.1	−0.1 / +0.0	+62.1 / +60.8	+41.0 / +45.0	+30.6 / +26.6
ULIP-2 KID, *cut*	+9.8 / +6.4	+2.2 / +1.3	+52.5 / +41.8	+34.3 / +30.8	+22.9 / +15.0
chamfer MMD, *all*	+6.3 / +8.0	+1.1 / +1.7	+55.0 / +66.2	+13.1 / +15.9	−0.0 / +0.2
chamfer MMD, *vis*	−0.1 / −0.1	−0.4 / −0.3	+64.3 / +75.2	+8.0 / +9.5	−0.2 / −0.1
chamfer MMD, *cut*	+7.8 / +6.9	+1.8 / +1.4	+55.6 / +51.2	+14.9 / +14.0	+0.0 / −0.2
EMD MMD, *all*	+4.5 / +3.6	+1.0 / −0.0	+36.5 / +31.8	+11.2 / +8.8	−0.3 / −0.7
EMD MMD, *vis*	+0.0 / +0.6	+0.0 / +0.1	+40.7 / +45.9	+10.5 / +11.1	−0.0 / +0.1
EMD MMD, *cut*	+3.6 / +3.9	+0.4 / +0.4	+27.0 / +31.6	+8.9 / +9.5	+0.1 / −0.7

	ladder, n=128					board, n=256		
row	FILL-INTERIOR	HOLLOW-SHELL	MATERIAL-SHUFFLE	MONOCHROME	gates	FILL-INTERIOR	ρ V-KID	ρ MV-KID
V-KID	**+111.4**	**+71.0**	0	0	**pass**	**+444.4**	1	0.27
occupancy KID	+48.4	+13.1	0	0	G2, G6	+121.2	0.83	0.23
Uni3D KID, all	+26.6	+10.4	+22.4	+99.7	G6	+31.8	0.51	0.85
Uni3D KID, vol	+37.2	+10.6	+6.7	+105.5	G6	+108.4	0.59	0.78
ULIP-2 KID, all	+4.6	+1.8	+29.1	+101.8	G6 / G2	+12.0	0.42	0.87
ULIP-2 KID, vol	+11.9	+9.0	+9.7	+62.8	G6 / G2	+43.2	0.51	0.85
chamfer MMD, all	+7.6	+1.4	−0.1	−0.1	G2, G8	+8.9	0.64	0.27
chamfer 1-NNA, all	+2.9	+1.1	0.0	−0.6	G2, G6	+4.0	0.76	0.43
EMD MMD, all	+3.2	+0.3	−0.2	−0.3	G2, G8	—	—	—
EMD 1-NNA, all	+1.6	+0.1	−0.5	−0.2	G2	—	—	—
chamfer MMD, voxel centres	−3.4	+2.8	+0.1	+0.1	G2, G8	−3.5	0.53	0.54
IoU MMD, D4-aligned	−9.0	+7.7	0	0	G8	−5.3	0.36	0.35
IoU 1-NNA, D4-aligned	+2.7	+2.4	0	0	G2	+5.9	0.75	0.28
IoU MMD, corner	−11.5	+8.4	0	0	G6, G8	−7.4	0.52	0.46

Table 13. **V-KID against structural alternatives.** Separation from held-out real data in each row's own noise floors. Gates: G2 noise ordering, G6 invariance, G8 n-stability; "G6 / G2" gives the failing gate under the geometry and the render rungs. ρ: Spearman correlation of the ordering of eleven leaderboard arms. The 1−coverage rows fail at least two gates each and are omitted. Exact EMD runs on the ladder only. The ladder scores against its 248-build subset reference and measures from the null mean, so V-KID reads +111.4 there and +201.8 in the protocol run at the same n.

grid of exterior views)." for exterior pairs, and for cutaways states that the upper half of the build, or in each view the corner facing the camera, has been removed so the reader can see inside. The registered predictions were chance for both interior probes on exterior views, above chance for both on cutaways, certainty against RANDOM-VOXELS, and a preference for GABLED-HOUSE over PATCH-STITCH.

Table 14. Probability the judge prefers the first arm, with cluster-bootstrap 95% intervals. Every pre-registered prediction holds except the last, where the judge agrees with both metrics.

pair	view	n	P(first)
real vs FILL-INTERIOR	exterior	256	0.53 [0.50, 0.56]
real vs FILL-INTERIOR	roof off	256	0.70 [0.64, 0.75]
real vs FILL-INTERIOR	quadrant cut	128	0.63 [0.53, 0.72]
real vs HOLLOW-SHELL	exterior	256	0.52 [0.49, 0.54]
real vs HOLLOW-SHELL	roof off	128	0.73 [0.66, 0.80]
real vs RANDOM-VOXELS	exterior	128	1.00 [1.00, 1.00]
GABLED-HOUSE vs PATCH-STITCH	exterior	256	0.28 [0.21, 0.35]

On exterior views of a house and its filled twin the judge picks the first slot 93% of the time and the two orders agree on only 13% of pairs, so its preference there is position, not content. Houses whose fill shows through a window do

Table 15. Descriptor ablation, n=128, 48 real draws per variant. Each separation is in that variant's own noise floor, measured under that variant. The two suspect terms are the explicit size scalar and the sealed-interior scalar; deleting either or both leaves the separation intact.

descriptor variant	dims	floor	FILL-INTERIOR	GROW-SHELL
full	77	0.1097	**201.8**	214.5
deleting a suspect term				
without `log1p_n_blocks`	76	0.1114	202.2	214.9
without `interior_ratio_sealed`	76	0.1127	197.1	211.7
without both	75	0.1147	197.3	211.9
deleting a whole block				
without pattern bins	22	0.5113	198.0	120.9
without wall thickness	72	0.1163	176.4	216.6
without height profile	69	0.1312	199.6	213.9
without global scalars	68	0.1302	171.6	206.7
one block alone				
pattern bins only	55	0.1802	113.9	201.3
wall thickness only	5	6.0167	158.7	75.7
height profile only	8	1.6293	8.0	28.9
global scalars only	9	1.7594	164.5	118.1

not move it (0.53 against 0.50 for the rest). At the system level a Bradley–Terry fit over eleven arms ranks held-out real sixth, with FILL-INTERIOR and both VOXEL-AR arms above it. Its Spearman agreement with the n=256 leaderboard is 0.74 for MV-KID, 0.54 for BlockScore and 0.20 for V-KID (0.76, 0.57 and 0.16 at n=128), over 3D-Craft subsets for the judge and full scored sets for the metrics. The judge reads exterior views and shares the render metric's null space, so we report it as a third exterior observer rather than as ground truth.

J. Robustness of V-KID

A hand-designed descriptor invites the objection that its features were chosen to catch our probes. We answer it at two sample sizes: by deleting the two suspect terms at the protocol run's n=128, and by removing whole families, drawing random subsets and replacing the descriptor with generic maps at the leaderboard's n=256.

J.1. The two suspect terms

The two objections are that V-KID separates FILL-INTERIOR only because the probe adds mass and the descriptor carries an explicit size term, or only because the descriptor carries a sealed-interior scalar the probe drives to zero. Table 15 deletes both terms, alone and together, and each block of the descriptor in turn, with 48 fresh real draws per variant, each separation in that variant's own noise floor. Neither term carries the result, and the mass-matched GROW-SHELL column separates as strongly as FILL-INTERIOR.

J.2. Families, random subsets and generic descriptors

Each test below runs under the bench's own scoring code: the blocking gates on the ladder (n=128, 48 fresh real draws, geometry rungs) and the separations on the n=256 leaderboard of record, whose build hashes we verify. Every row is whitened on the reference and scored by the same unbiased KID, and every separation is in that row's own noise floor. Rank agreement is Spearman's ρ over the nine structural arms of the board against full V-KID.

No family carries the result. We group the 77 dimensions into seven families: 55 pattern orbits, thickness (5 bins and 2 scalars), height (8 bins and the entropy), surface (2), symmetry (1), interior (2) and size (1). Removing any one family keeps every gate passing, FILL-INTERIOR between 310 and 457 floors (full 444.4), HOLLOW-SHELL between 146 and 270 (full 265.4), and $\rho \geq 0.95$ (Table 16). Random subsets tell the same story without any grouping of ours: 20 of 20 random 38-dimensional subsets pass every gate, with FILL-INTERIOR at a median of 351 floors (minimum 283), HOLLOW-SHELL at 221 (133) and ρ at 0.97 (0.88). At 19 dimensions, 17 of 20 pass, and the weakest still separates FILL-INTERIOR by 179 floors.

A descriptor with no hand design passes. We replace the descriptor by generic maps ϕ of the occupancy grid, made exactly invariant to the yaw group by orbit averaging, $f(x) = \frac{1}{8}\sum_{g\in D_4}\phi(gx)$, which is the only design step they share. Three are random-weight three-layer 3D CNNs with mean and max pooling per layer (224 dimensions, three weight draws), one is a feature-hashed histogram of raw 3^3 occupancy patterns (256), and one counts occupied 3^3 neighbours of solid and air cells at $1\times$, $2\times$ and $4\times$ pooling (168). All six rows, including the concatenation of a CNN with the counts, pass every gate and separate FILL-INTERIOR by 257 to 437 floors and HOLLOW-SHELL by 166 to 297. Occupancy KID on a centred, pooled grid fails noise ordering and invariance (Sec. I.2); orbit averaging is what turns the raw volume into a valid metric, and the separation follows from observing the volume rather than from the choice of features. The generic rows reproduce V-KID's ordering of probes and baselines (ρ between 0.8 and 1.0 over five arms) and differ from it in the order of the four learned arms, where the random CNNs place Dream-Cubed behind the three autoregressive arms; we report V-KID as the instantiation, and the ordering of close learned arms as descriptor-dependent.

Global coherence is a separate axis. PATCH-STITCH tiles 8^3 patches of different real builds, so it is the natural attack on a local descriptor. V-KID places it 368.2 floors

Table 16. V-KID robustness. Gates on the ladder (n=128); separations on the n=256 board in each row's own noise floor; ρ: Spearman over nine arms against full V-KID. Random rows give the median and, in brackets, the minimum over 20 draws.

descriptor	dims	gates	FILL-INTERIOR	HOLLOW-SHELL	PATCH-STITCH	ρ
full V-KID	77	pass	444.4	265.4	368.2	1.00
leave one family out						
without pattern orbits	22	pass	457.4	146.0	342.1	1.00
without thickness	70	pass	310.4	265.8	374.4	0.95
without height	68	pass	414.0	251.4	342.7	1.00
without surface	75	pass	441.8	258.3	360.9	1.00
without symmetry	76	pass	451.1	270.3	357.7	1.00
without interior	75	pass	421.8	258.7	327.0	1.00
without size	76	pass	440.9	263.8	374.6	1.00
random subsets, 20 draws						
50%	38	20/20	351 [283]	221 [133]	317 [230]	0.97 [0.88]
25%	19	17/20	298 [179]	153 [56]	246 [178]	0.95 [0.85]
no hand design, orbit-averaged						
random 3D CNN, draw 1	224	pass	339.4	223.8	277.8	0.80
random 3D CNN, draw 2	224	pass	437.3	297.4	284.7	0.73
random 3D CNN, draw 3	224	pass	403.6	242.1	309.5	0.80
hashed 3^3 patterns	256	pass	256.7	208.9	326.0	0.75
multi-scale neighbour counts	168	pass	359.3	166.3	514.9	0.80
random 3D CNN + counts	392	pass	396.4	230.7	406.1	0.78
global terms						
V-KID + connected parts	79	pass	454.6	271.4	366.1	1.00
V-KID + pooled orbits	187	pass	485.3	253.4	354.1	1.00
connected parts alone	2	pass	0.0	0.1	258.4	−0.20

from real, fourth of nine arms. Appending global terms, the largest 26-connected part ratio and component count, or 2^3 orbit histograms of the $2\times$ and $4\times$ max-pooled grid, keeps every gate and leaves it fourth, with its separation between 0.73 and 0.81 of FILL-INTERIOR's (0.83 for V-KID). The connected-part ratio on its own separates PATCH-STITCH by 258 floors and both interior probes by none, so coherence is measurable and orthogonal to the interior axis; at two of 79 dimensions it does not move the joint kernel. We keep V-KID unchanged and state its scope as local geometry.

J.3. Kernel and preprocessing

With unit-norm rows and $t = a^\top b/D \in [-1/D, 1/D]$, the cubic kernel expands as $(1+t)^3 = 1 + 3t + 3t^2 + t^3$. On every probe the linear term carries 99.8% of MV-KID and 98.8 to 99.5% of V-KID, so both metrics are in effect tests of a difference in mean embedding; differences in spread, such as mode collapse, are left to the diversity and memorization gates. Table 17 varies the kernel and the preprocessing. Every V-KID variant leaves FILL-INTERIOR and HOLLOW-SHELL hundreds of floors from real, scores MATERIAL-SHUFFLE at exactly 0 and passes every gate of the geometry ladder at n=64 and 128. Whitening and the clip set the magnitudes: removing the clip lowers FILL-INTERIOR from 422 to 288 floors, and removing whitening raises it to 1,637 while lowering GROW-SHELL to 165. On the render side, whitening the DINOv2 features against the reference leaves FILL-INTERIOR at 0.50 floors and HOLLOW-SHELL at 0.06.

Per-build disagreement on real builds. The two tiers disagree about real, uncorrupted builds as well, with no probe involved. The unbiased estimator is a sum over builds, so ranking the same 128 held-out builds by each metric's exact per-build contribution gives two nearly independent orderings, and only the geometric one tracks how much interior a build has (Fig. 9). Nearest neighbours dissociate the same way (Fig. 8).

Table 17. **Kernel and preprocessing ablation.** Separations in each variant's own noise floor (the bench estimator's expected value, so the base V-KID row reads 422 where the card's 48-draw floor gives 444), n=256, 399-build reference. Gates: the geometry ladder at n=64 and 128; *the render ladder at n=64.

variant	FILL-INTERIOR	HOLLOW-SHELL	GROW-SHELL	MATERIAL-SHUFFLE	gates
V-KID, cubic	422	252	446	0	pass
linear kernel	419	251	444	0	pass
RBF, median	475	266	485	0	pass
no clip	288	247	409	0	pass
no whitening	1,637	213	165	0	pass
no whitening, RBF	692	160	180	0	pass
MV-KID, cubic	0.31	0.03	206	365	pass*
linear kernel	0.31	0.03	206	365	—
RBF, median	0.32	0.03	215	351	pass*
whitened DINOv2	0.50	0.06	200	277	—

K. LEGO and BrickGPT

BrickGPT [23] is trained on the StableText2Brick (ST2B) training split, so the paper's pooled LEGO corpus is not a fair reference for it. We build the reference from the 4,785 ST2B *test* structures alone, with no void filter; ST2B test objects share no object with its training split. The suite's group split divides them 3,349/718/718 and the reference is 512 of the 718 validation structures. We prompt with one caption of each of 256 structures drawn from the suite's test partition, so the prompted objects are in neither the reference nor BrickGPT's training data. Generation uses the released configuration with three forced deviations: bf16 weights, a mirror of the base model with identical size, and the repository's documented connectivity rollback in place of the physics rollback, because the free Gurobi licence refuses every ST2B structure. A 1,800 s cap per prompt (not in BrickGPT) produced 3 failures. Stability is solved with SCIP on BrickGPT's own formulation, validated against the Gurobi scores ST2B ships (240 of 240 agree, mean difference 4.6×10^{-7}).

Table 18. BrickGPT's own metrics. Stability is over structures the solver finished.

	valid	stable	mean stab.	min stab.
paper, physics rollback	100%	98.8%	0.996	0.915
paper, no rollback	100%	24.0%	0.947	0.228
ours, connectivity rollback	**98.8%**	**95.8%**	**0.995**	**0.896**
held-out real (same 256)	100%	100%	0.997	0.953

Table 18 places our run close to the published full method. The three-point stability gap is what replacing

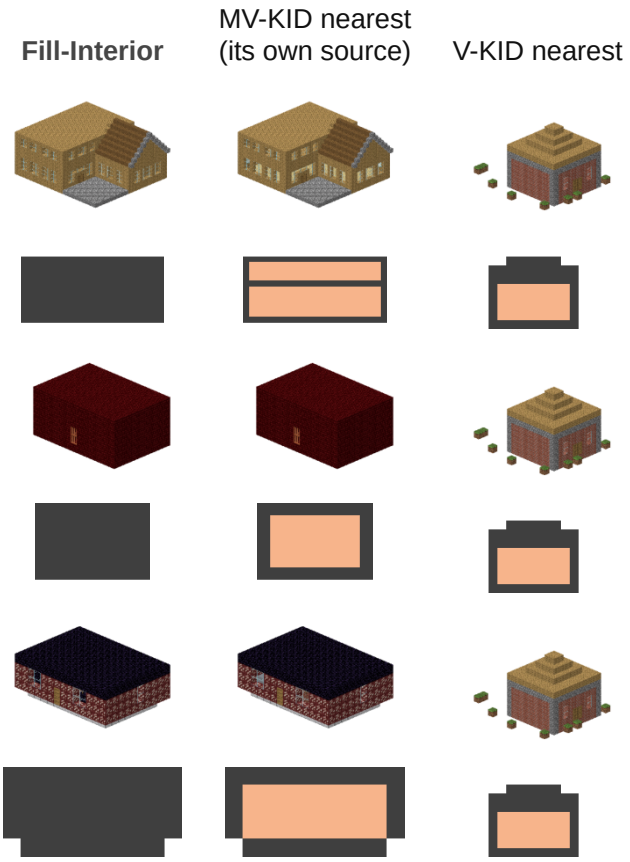


Figure 8. **Each feature space is blind where the other sees.** Three filled 3D-Craft houses (left), the nearest held-out real build in MV-KID's feature space (middle) and in V-KID's (right), each over a cross-section with sealed air in peach. The render features return the house's own source for every one of 256 filled houses; V-KID's features return it for 26% (median source rank 17.5) and send a filled house to a few dense hub builds, the same one for all three rows. For MATERIAL-SHUFFLE the pattern reverses: V-KID returns the source every time and the render features 26%. For held-out real builds the two spaces pick the same nearest reference build 3.9% of the time (chance 0.25%).

the physics rollback by the connectivity rollback should cost, since a connected but cantilevered structure passes the connectivity check. On the suite (n=253, 77-dimensional descriptor), BrickGPT sits at +11.77 on appearance and +51.48 on geometry, BlockScore 51.48, behind the solidified control at 36.41 (−0.09 appearance), and the held-out structures matched to its prompts score like real (0.40). Its largest-component ratio is 0.980 against 0.999 and its sealed interior 0.011 against 0.034, so part of its geometric gap is interior volume the render tier cannot see. It is not disqualified on novelty. Fig. 10 shows the domain in its own colors.

L. Leaderboard Arms and Real Generators

This section documents the one external submission, the contamination case study and the blindness on 40 real generators. Fig. 13 shows six builds of every leaderboard arm.

L.1. Dream-Cubed

Dream-Cubed [18] is a 279M-parameter 3D diffusion transformer over 32^3 Minecraft chunks, conditioned on one of 15 biome classes and trained on procedurally generated worlds that contain no GrabCraft, 3D-Craft or text2mc build. We run the `MD4_p4_boosted` checkpoint (Apache-2.0), conditioned on `village`, with the configuration's own classifier-free guidance of 4.0 and 1,000 reverse steps, for 288 chunks; the leaderboard scores the first 256 extracted builds.

Extraction rule. A sample is a village on terrain. We keep the constructed blocks; count sandstone only where no terrain block lies above it in its column, since desert subsurface is sandstone under sand; select the 26-connected component with the most non-log blocks, so a bare tree never wins; crop it to its bounding box; and drop chunks whose component has fewer than 80 blocks. 284 **of** 288 **chunks yield a build.** The rule is ours and favours the model, so the whole chunk is scored beside it as a control: every chunk is a 32-wide slab of stone and dirt, and the diversity gate disqualifies it for mode collapse at +24.0 spreads. Two vocabulary losses come from Dream-Cubed itself: its slab and stair tokens carry no material (mapped to stone slab and oak stairs), and its block names carry no variants (planks, logs and wool map to their first variant). Both raise its palette pillar, which reads +87.65 and never determines its BlockScore.

Novelty against the training split. Against the houses_32 training split the extracted builds reach a mean nearest-neighbor voxel IoU of 0.42, against 0.38 for held-out real, and one of 256 (0.004) is a duplicate at IoU 0.95, against 0.008 for held-out real. That gate compares against our training split only.

Within-set near-twins, and the gate they prompted. The builds repeat one another. For each build we take its nearest other build in the same set, at the best occupancy IoU over the 8 yaw and mirror variants and every integer translation (exact, by FFT cross-correlation). 59.8% of Dream-Cubed's 256 builds have such a twin at IoU $\geq$ 0.8, against 0.8% of held-out real, and the median nearest-twin IoU is 0.85 against 0.42. The largest group among all 284 extracted builds holds 28 copies of one 117-block hut. Size does not explain it: restricted to Dream-Cubed's block-count interquartile range and matched at 98

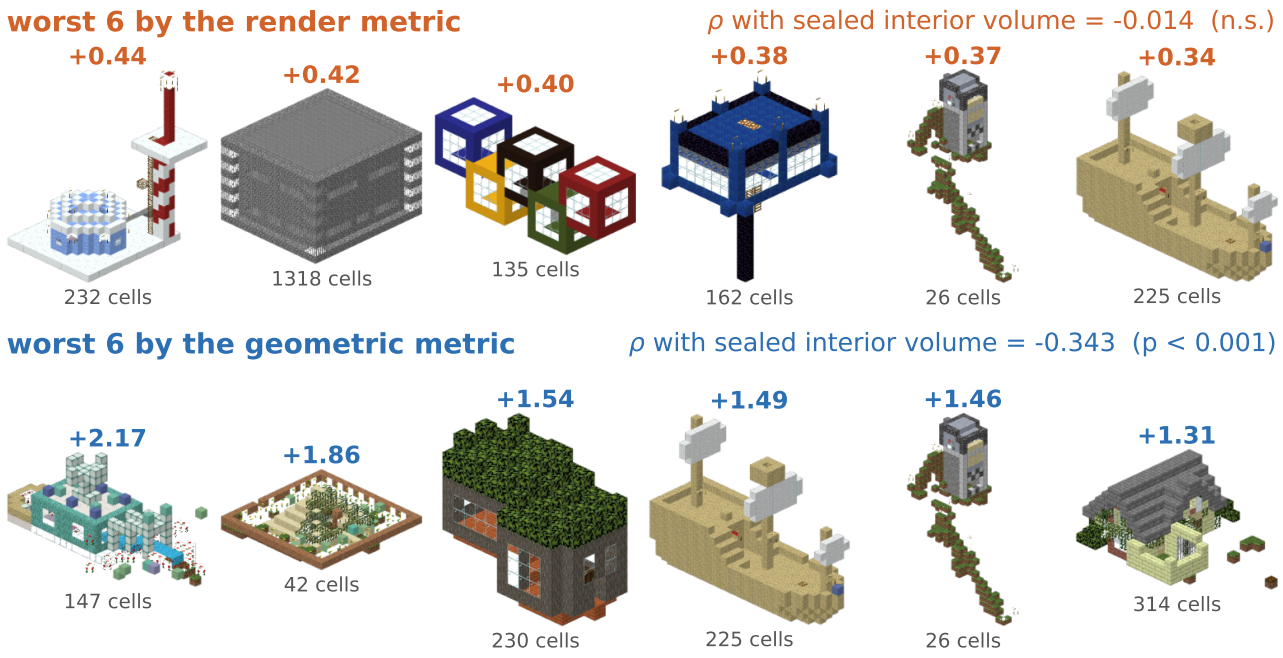


Figure 9. **The two metrics disagree about real builds, with no probe involved.** The unbiased estimator is a sum over builds, so each build's contribution to its arm's score is recoverable. The six builds each metric penalizes most are shown with their contribution and their sealed interior volume. The two orderings are nearly independent, $\rho = +0.167$, and only the geometric ordering tracks how much interior a build has: its rank correlation with sealed interior volume is -0.343 ($p < 0.001$) against -0.014 for the render metric. Builds shown are 3D-Craft only, for licensing.

builds per set, the rate is 73% against 0%. Neither existing gate sees this. The duplicate gate compares to the training split, and the diversity gate averages over all pairs, which 160 clusters of near-copies still pass (0.866 against 0.898 for real). We therefore added a third novelty gate, the *within-set near-twin rate*, under the same rule as the duplicate gate: an arm is disqualified when its rate lies more than 3 spreads above the real control of the same run, with a spread of at least 0.02. At n=256 the bar is 0.068. Fresh real draws of 256 sit well below it (0.006, 0.008 and 0.019 on average from the test, validation and training splits). The gate judges submissions, not diagnostics: probes that rewrite real builds are exempt, since filling interiors makes real houses genuinely more alike (0.066 for FILL-INTERIOR), while the repeated-build control stays judged. Every model we trained reads 0.000. The gate disqualifies Dream-Cubed at +29.5 spreads, and with it the two parametric baselines, HOLLOW-BOX (0.246) and GABLED-HOUSE (0.207), whose shapes repeat with different materials. It reads occupancy only.

What the builds are, and the templates control. Of the 284 extracted builds, 45% are houses with a room, 36% are roofless enclosures and 19% are fragments, against 99% houses in held-out real, at a median of 272 blocks against 675 for the reference. Size carries at most 7% of the geometric distance: refitting the standardizer without the $\log(1+n_{\text{blocks}})$ column moves raw V-KID from 7.67 to 7.17, and scoring against only the 111 reference builds inside Dream-Cubed's own block-count interquartile range (155–462) raises it to 8.57. Their median best IoU to the 299 village templates of the current game version is 0.52, against 0.32 for held-out real, and none reaches 0.9. As a control, we pass the current templates through Dream-Cubed's own extraction and score them under the leaderboard's protocol: 174 house templates survive the 80-block floor (183 village templates in all). Their pillars would give a BlockScore of 202.95 (194.98 for all 183), geometry-limited, ahead of every model we trained, from a set that involves no learning. They repeat each other as well (67.2% twins), and the near-twin gate disqualifies them at +33.2 spreads. Scored in the same run, Dream-Cubed reproduces its leaderboard values exactly.

L.2. Case study: contamination no statistic sees

The main results use corruptions we built, where ground truth is known by construction. This case study asks whether the suite catches a failure we did not build. A gen-

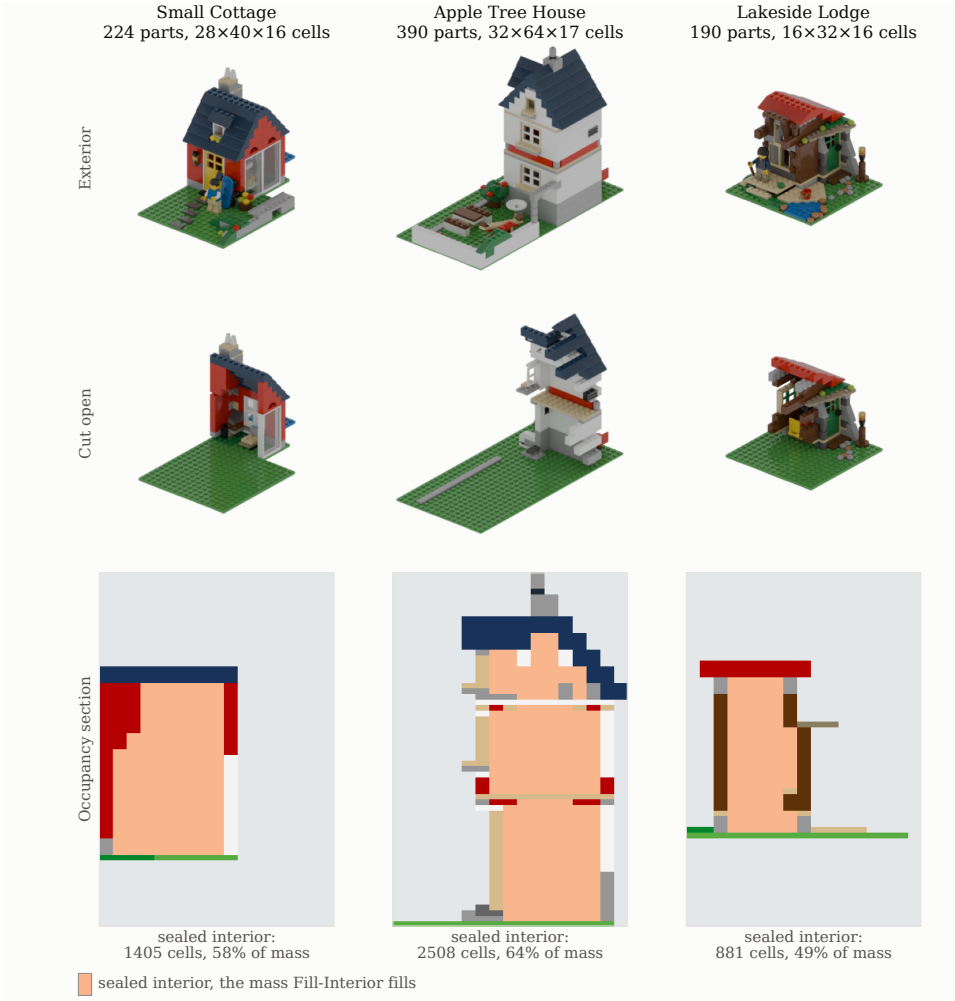


Figure 10. **A second domain, with real colors.** Three LDraw.org OMR models in their own LEGO colors: the exterior an exterior protocol sees, the same parts cut past one plane, and the occupancy grid we derive sliced at that plane, sealed interior in peach, every other color the parts' own, all renders at one scale. This domain shares only the fact that builds enclose space: different parts, different lattice, different renderer, 190–390 parts against thousands of blocks. The measured replication of main paper §6.3 runs on StableText2Brick rather than on these three models, which we picture because their terms permit it (Sec. M). Official LEGO sets are far less hollow than either measured corpus: among models with at least eight sealed cells, the basis the OMR audit reports, FILL-INTERIOR adds a median $+1.2\%$ mass to an OMR model against $+35.8\%$ to a held-out house. Models: Small Cottage (Stefan Frenz), Apple Tree House (Marc Giraudet), Lakeside Lodge (Stefan Frenz); LDraw.org OMR, CC BY 2.0, modified only by the deletions shown.

erator we control is the right subject, because its training split is knowable. The suite requires every arm to declare that split. VOXEL-AR, the top-ranked submission of the protocol run, declared one that contained 346 of the 399 canonical test builds (87%). LLM-FT and LLM-FT-GEO declared an earlier 90/10 shuffle holding 53 of the 399 (13%). No distribution statistic sees either. In the protocol run, VOXEL-AR's duplicate rate is 0.000 and its retrieval percentile 0.509 against a real held-out floor of 0.534, far from the verbatim-copy control at $+10.7$ spreads ($+10.1$ on the n=256 leaderboard).

Retraining VOXEL-AR on the canonical split, with architecture and hyperparameters unchanged, isolates the effect (Fig. 11). We report raw V-KID, the statistic the permutation test of Sec. D.4 compares; all four arms were scored together in the protocol run at n=128, against one reference. Contaminated, the arm scores 11.62 and separates from one language-model baseline at Holm 5.5×10^{-3}; clean, it scores 14.898 against their 15.006 and 15.869 and separates from neither. Under a correctly calibrated test it never held a significance group alone (Table 19). Contamination lowers an arm's distance from real, so the baselines'

Table 19. Retraining on a clean split collapses the margin. V-KID at n=128 (protocol run), 399-build reference, paired and Holm-corrected over all 120 pairs with an exact label-permutation test. Letters mark maximal sets of arms that no test separated at $\alpha = 0.05$: a shared letter means the pair was *not separated*, never that the arms are equivalent, and an arm carries several letters when non-separation fails to be transitive, as it does here. The contaminated arm is not separated from either its own clean retrain or the stronger language-model baseline, and the clean retrain ties with both baselines it had appeared to beat. VOXEL-AR-U's memorization gate is marginal here (+3.0 spreads against a bar of 3.0) and does not fire at n=256 (Sec. D.3).

	arm	V-KID ↓	group	status
	REAL (floor)	−0.095	a	—
	TRAIN-COPY	0.960	b	DQ memorization
submissions	VOXEL-AR† *(contaminated)*	11.624	**c**	unranked
submissions	**VOXEL-AR** *(clean)*	14.898	**cd**	ranked
submissions	LLM-FT	15.006	**cd**	ranked
submissions	LLM-FT-GEO	15.869	**d**	ranked
submissions	VOXEL-AR-U *(clean)*	19.570	e	**DQ memorization**
	PATCH-STITCH	19.976	e	ranked
	RANDOM-VOXELS	37.729	h	ranked

own contamination flatters them and we report that comparison as not separated. The sibling VOXEL-AR-U, retrained on the same clean split at a duplicate rate of exactly 0.000, sits on the memorization bar: at n=128 its retrieval percentile of 0.382 against a real floor of 0.534 is +3.0 spreads, which the gate disqualifies, and at n=256 it reads 0.403 against 0.507, +2.1 spreads, and ranks fourth. A gate decision this close to its bar is a question of sample size, which is why the protocol asks for $n \geq 256$.

The language-model arms. LLM-FT and LLM-FT-GEO appear only in the protocol run's case study, not on the leaderboard. They decode at temperature 0.8 under a 1,200-token cap, and that cap limits both arms: 11 of 128 generations came back empty, 139 attempts were needed for 128 samples, and several completions are truncated into structures one block thick.

What detected both contaminations was a required field rather than a detector: every memorization gate passed and the declaration did not, so benchmarks in this domain should require that declaration and refuse to rank arms that cannot make it.

L.3. The blindness on real generators

The probes of the main text alter real builds. This section asks whether the same blindness changes conclusions about generators. We pool every house generator scored against the 399-build validation reference at the protocol's n under the current renderer: 22 autoregressive voxel models (scale, temperature, best-of-k, rejection fine-tuning, tokenizer and schedule variants), 10 facade generators, which emit the six face images of a build and lift them into the voxel grid, 5 LLM program generators, the two clean 3D-BPE models of the leaderboard and BrickNet [13], 40 in all, 35 of them at n=256 and the LLM arms at n=128 to 133. Both metrics are recomputed from the per-build features of every arm, matched to its card by hash (MV-KID to 3×10^{-6}). Every pair is tested with the exact label-permutation test of Sec. D.4 against the fixed reference, 20,000 permutations, Holm-corrected over all 780 pairs per metric. The test stays exact at unequal sizes, and the 35 arms at n=256 are also reported alone. The interval on ρ half-samples every arm and the reference jointly.

Table 20. **Agreement of the two rankings over real generators.** Spearman correlation of the MV-KID and V-KID orderings with a half-sampling 95% interval. Pairs: separated by both metrics under the exact permutation test after Holm correction over all 780, and the subset the two metrics order oppositely.

generators	k	ρ	both sep.	opposite
all	40	0.50 [0.37, 0.58]	308	**40**
at n=256	35	0.60 [0.46, 0.67]	255	24
autoregressive voxel	24	0.37 [0.16, 0.51]	58	7
facade	10	0.99 [0.88, 1.00]	32	0

Where the rankings part. The interval of Table 20 reflects sampling of builds. Resampling the generators themselves widens it to $[0.20, 0.72]$, and a cluster bootstrap over the five families to $[0.02, 0.91]$: the correlation is a property of this pool's family mix. Without the facade family it is 0.28; without the voxel family, 0.72. In 22 of the 40 reversed pairs the render metric prefers a facade generator. Facade generators emit shells: a median 0.5% of their blocks have no face on exterior air, against 15.6% for held-out houses and 12.5% for the voxel models, and moving from the MV-KID to the V-KID ranking costs the family 7.7 places on average. Across all 40 generators, however, the shift between the two rankings correlates with the share of hidden blocks only at $\rho = -0.26$ (p=0.10), so hidden mass alone does not explain the reordering. The pair of Fig. 12 is tied under MV-KID, 3.1 noise floors apart (Holm p=1), and 292.9 floors apart under V-KID (Holm p=0.039, the smallest value 20,000 permutations can reach over 780 pairs); with the roof removed, the facade builds are walls around an empty floor. Next to the same voxel model, a best-of-four facade generator is preferred by MV-KID by 55.1 floors and rejected by V-KID by 209.5, both at the same Holm floor.

The ceiling of a shell generator. Reducing each held-out house to the cells its six faces show gives the output of a perfect facade generator. It sits 8.1 noise floors from real

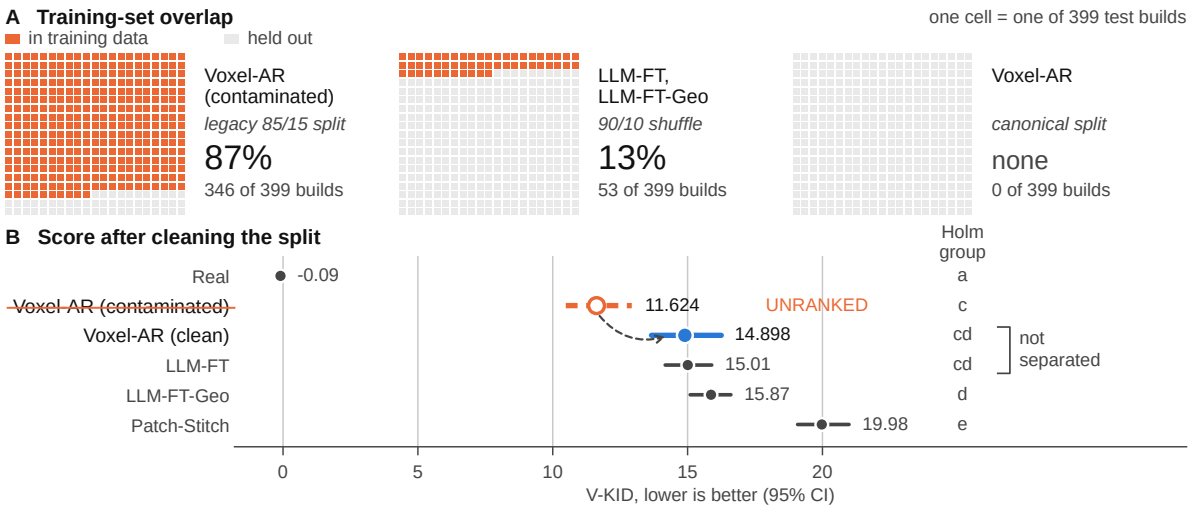


Figure 11. **Declared provenance, and what it cost.** *A:* how much of the canonical test split each arm had already trained on; the third waffle is VOXEL-AR retrained on that split. *B:* the retrain costs $+3.27$ raw V-KID and erases the margin. No memorization gate fires on the contaminated arm, so it is unranked rather than disqualified.

data under MV-KID, ahead of every trained generator by more than 90 floors, and at $+310.6$ under V-KID, 19th of 41. Applied to generator output, HOLLOW-SHELL deletes the blocks no camera sees from our best voxel model, 15% of its mass, and moves it from 5th to 29th of 41 under V-KID ($+147.5$ floors). On the same 128 builds, rendered and scored together with their hollowed copies, MV-KID moves $+0.15$ noise floors and V-KID $+117.1$; for the 3D-BPE model the two read $+0.18$ and $+139.2$. The converse holds as well: a voxel model sampled at temperature 0.7 is over-solid, 24% hidden blocks, and hollowing its output moves it 24.7 floors *closer* to real data under V-KID. V-KID therefore sees what is deleted from a generator's interior; whether what it sees is damage is a separate question, which Sec. L.4 answers.

Design decisions. We list 34 decisions taken while building these generators, each a pair of arms that differ in one choice (temperature, best-of-k, rejection fine-tuning, scale, D4 augmentation, tokenizer, schedule, LLM loop), and test each with the same exact permutation test, Holm-corrected over the 34. The two metrics agree on 6 and never disagree in sign when both are significant; 13 are significant under neither. On 15 the verdict depends on the pillar: 14 are significant under MV-KID alone, among them D4 augmentation of the 20M facade model (-52.4 floors under MV-KID, $+16.9$ under V-KID), and 1 under V-KID alone, replacing voxels with 3D-BPE pieces ($+162.6$ floors worse under V-KID, $+21.5$ under MV-KID). Most decisions that change a score change what the exterior shows; local geometry moves beyond noise in 7 of the 34.

Scope. The 40 generators come from one project. V-KID is a distribution distance on local geometry, not a counter of hidden blocks: the LLM program arms are also near-shells and move up, not down, under V-KID, and the best facade generator has the lowest V-KID of the pool, below its own ceiling, because sampling at temperature 0.7 narrows its size and shape distribution toward the reference mode. Under the 80-dimensional descriptor of the current cards, 29 of the 308 both-separated pairs reverse and the correlation rises to 0.65 $[0.54, 0.71]$; every statement above holds.

L.4. Detection is not ranking: a functional ground truth

The probes establish that V-KID detects a change to a real interior that MV-KID cannot see. On the 40 reversed pairs the question is different: which ranking orders generators by how usable their interiors are. We answer it with a test that reads neither metric's features.

A walk-through test. Every build is set on flat ground and entered by the player of the traversability probe, a body two blocks tall that steps up one block, falls, climbs ladders and opens doors. Five measures follow: whether the player reaches any interior standing cell (*enterable*); how many interior standing cells it reaches (*floor*); how many rooms it reaches, a room being a component of interior air holding at least six standing cells, with doorways closed so that a doorway separates two rooms; whether at least half of the plan is roofed by a block with two free cells beneath it; and the share of blocks in the largest 6-connected component. Held-out houses average 0.94, 80 cells, 1.36 rooms, 0.81 and 0.98. V-KID has no reachability, door se-

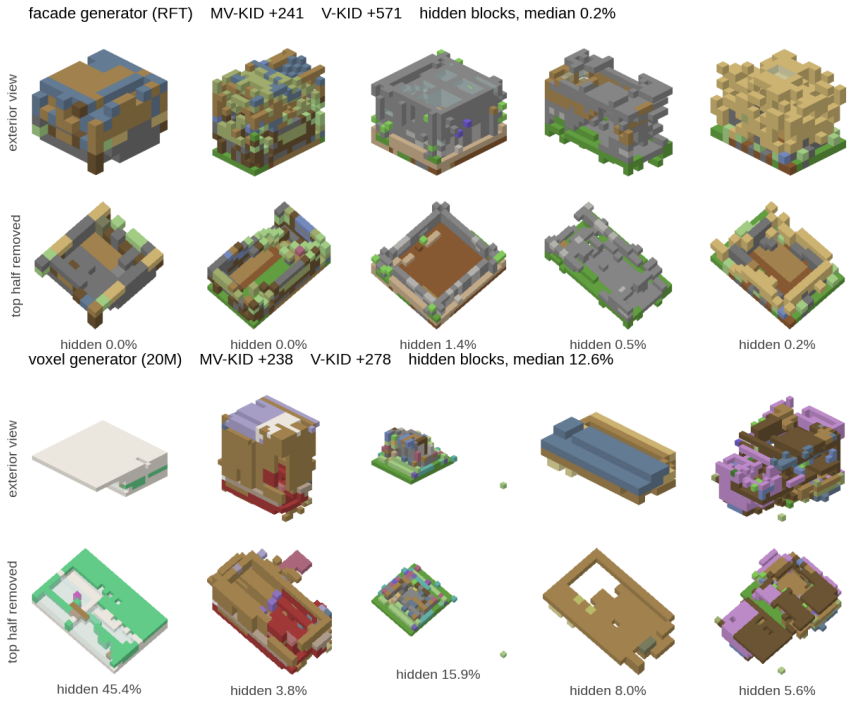


Figure 12. **Tied under MV-KID,** 293 **floors apart under V-KID.** The first five builds of each arm that pass the memorization gate, generated output only. Top of each block: the exterior corner view of the protocol; bottom: the same build with its upper half removed. The facade generator's builds are walls around an empty floor, a median 0.2% of blocks hidden from every exterior ray; the voxel model's hold partitions and fill, 12.6% (15.6% for held-out houses).

mantics, room labelling, plan shelter or component analysis; a cross-validated ridge regression on its 77 features explains $R^2 = 0.29$ to 0.62 of the per-build measures. The functional distance FD of an arm is the mean over the five measures of the Wasserstein-1 distance to the validation reference, each in units of the real between-build spread, with 1,000 bootstrap replicates that share one reference resample. Three variants check the construction: units of real-versus-real sampling distance at n=256, a cubic-kernel KID on the standardized measures, and an occupancy-only version without door semantics.

The test sees interiors. Held-out real data ranks first of 41 under all four variants, at FD 0.09 $[0.08, 0.19]$ against 0.52 for the best generator. FILL-INTERIOR reaches 1.21, cutting the enterable share from 0.94 to 0.20, and HOLLOW-SHELL 0.60: the walk-through detects the two probes MV-KID cannot, so its verdict on generators is not a second exterior observer.

On generators it follows the render ranking. Over the 40 generators, FD correlates with MV-KID at Spearman 0.67 $[0.41, 0.82]$ and with V-KID at 0.16 $[-0.20, 0.45]$, with intervals that resample generators and FD replicates jointly. The maximum of the two, which BlockScore reads, correlates at 0.17 $[-0.18, 0.45]$, because geometry is the worse pillar for 37 of the 40. On the 40 reversed pairs, every FD difference is significant, and the walk-through sides with MV-KID in 33 and with V-KID in 7 (18%, Clopper–Pearson $[7, 33]$). The facade family does not produce this. On the 18 pairs without a facade generator it sides with V-KID in none ($[0, 19]$%); on the 22 facade pairs it does so in 7, five of them BrickNet preferred over a facade generator, and BrickNet's builds are almost never enterable (0.8%). The variants agree (2, 9 and 8 of 40 for the noise-unit, KID and occupancy-only distances; none of the 18 non-facade pairs under the occupancy-only one), and leaving out any one generator keeps the share between 6% and 24%. Taken one measure at a time, enterability, reached floor and reached rooms side with V-KID on none of the 40.

Why. The autoregressive voxel generators do not emit shells; they clutter their interiors. A median 22% of their builds can be entered, against 94% of held-out houses, with a median 3.4 reachable floor cells against 80. Their walls, block patterns and sealed air have the local statistics of real houses, and V-KID scores those statistics. Deleting the hidden blocks of the best voxel model moves it 147.5 V-KID floors away from real data and 0.41 functional units toward it (1.32 to 0.90), and the one design decision V-KID alone flags, 3D-BPE pieces over voxels, is one the walk-through prefers (1.34 against 1.67). Detection of interior change is not ranking of interior function. The probes hold, since FILL-INTERIOR and HOLLOW-SHELL are functional damage on real houses that renders cannot see. On current generators, though, the dominant interior failure co-varies with defects the render tier sees, and a local volumetric descriptor reads it as realism. A leaderboard that ranks by interior function needs a functional pillar beside both tiers.

Table 21. **Functional ground truth.** Top: functional distance to real (FD, 95% bootstrap interval) and enterable share. Bottom: on the 40 pairs the two metrics order oppositely, how often the walk-through agrees with each, and its Spearman correlation with each ranking over the 40 generators.

arm	FD	enterable
held-out real	0.09 [0.08, 0.19]	0.95
FILL-INTERIOR	1.21 [1.11, 1.32]	0.20
HOLLOW-SHELL	0.60 [0.52, 0.73]	0.95
best generator (LLM program)	0.52 [0.43, 0.65]	0.68
best voxel model under V-KID	1.32 [1.20, 1.43]	0.28
same, hidden blocks deleted	0.90 [0.78, 1.04]	0.59
	with MV-KID	with V-KID
reversed pairs, all 40	33	7
reversed pairs, 18 without a facade arm	18	0
reversed pairs, 22 with a facade arm	15	7
Spearman over 40 generators	0.67 [0.41, 0.82]	0.16 [−0.20, 0.45]

M. Reproducibility, Data and Licences

Released artifacts. The evaluation suite, the probes, the procedural baselines and the group-aware split are released and contain no upstream content. Every benchmark run writes a scorecard recording corpus, split hash, n, reference size, seed, host, device and commit, so each number in the paper is traceable to the run that produced it; the release README lists the runs of record and their cache manifests. **[TODO: Add the anonymized code URL and commit, and the institutional determination covering the corpus release, before submission.]**

Compute. Every run in the paper uses one workstation: one NVIDIA GeForce RTX 5070 Ti (16 GB), a 12-core AMD Ryzen 9 9900X and 32 GB of RAM. The n=256 leaderboard, rendering and embedding every arm and running the ladder and the head-to-head tests, takes 1,115 s end to end, and the n=128 protocol run 411 s. V-KID alone takes 15 to 18 s on one CPU process (Sec. I.2). Training VOXEL-AR takes about two hours on the same GPU.

Corpus and redistribution. Our Minecraft corpus derives from three public sources: the 3D-Craft house dataset [5, 7], released with the CraftAssist project; a scrape of `grabcraft.com`; and the text2mc dump, whose builds originate from `planetminecraft.com`. **We do not redistribute the GrabCraft- or text2mc-derived structures.** GrabCraft's terms permit sharing blueprints only by linking, and PlanetMinecraft's terms prohibit redistributing site data. We release instead the curation code, the per-build source URLs, content hashes, the group-aware split as index lists, and a rebuild script, so the corpus reproduces bit-exactly without our distributing third-party content. Builds are credited by source URL in the released manifests; the original scrape did not capture uploader names.

Licences. The 3D-Craft house data ships without a licence file and its distributing release, `facebookresearch/voxelcnn`, is CC BY-NC 4.0, so we treat it as non-commercial research material requiring attribution, not under the CraftAssist code's MIT terms. StableText2Brick and the BrickGPT code and weights are MIT-licensed; the dataset derives from ShapeNetCore, whose terms restrict use to non-commercial research. Dream-Cubed's village checkpoint and its data repository are Apache-2.0; we run its code for evaluation and do not redistribute it. The LDraw.org OMR models of Fig. 10 are CC BY 2.0.

Depiction. Every real build shown in this paper comes from **3D-Craft only**, since GrabCraft's terms name pictures of blueprints as a prohibited form of distribution; generated builds are shown as generated. Figures render builds with Minecraft's own texture assets, which are Mojang's copyright and which Mojang's usage guidelines permit in non-commercial images. The release contains **no texture files**, so a reader reproducing our figures supplies them from their own installation, and the figure scripts also produce flat per-block colors that need no permission. Minecraft features: © Mojang Synergies AB, included courtesy of Mojang AB. This is not an official Minecraft product and is not approved by or associated with Mojang or Microsoft.

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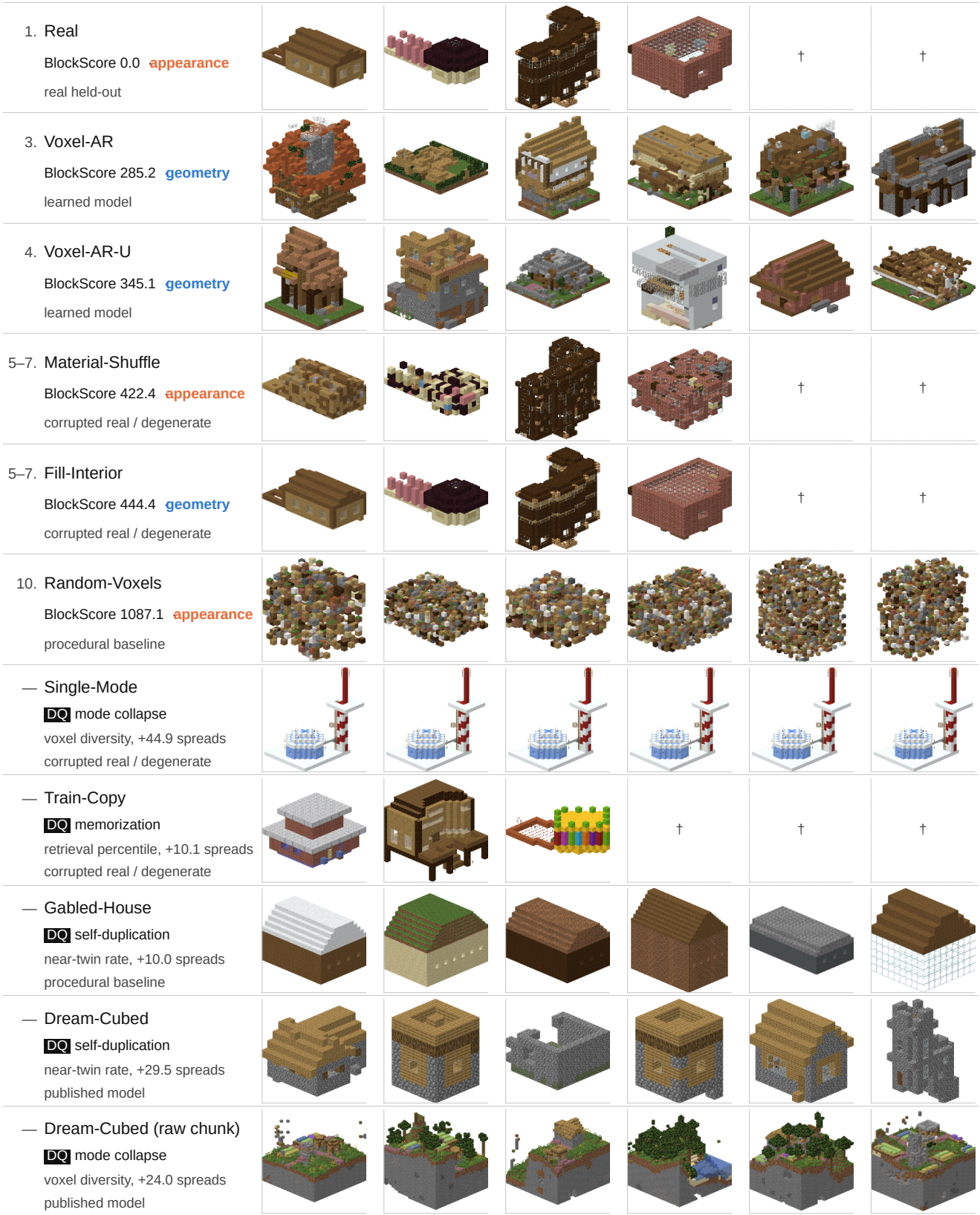


Figure 13. **Every arm, ordered by BlockScore** (n=256 leaderboard). Six builds per arm: the first six the run recorded, in the seeded order it recorded them, topped up from that arm's full scored sample where fewer than six remain. REAL, FILL-INTERIOR and MATERIAL-SHUFFLE are the same buildings column for column, so a corruption reads down a column and the paper's central claim can be checked by eye: the FILL-INTERIOR row scores +444.36 on geometry and +0.35 on appearance. Cells marked † are withheld under GrabCraft's terms, so the real-derived rows show four builds. Dream-Cubed is the extracted arm (Sec. L.1); its whole-chunk control is the last row. Each build is framed to its own bounding box, so cells are not to a common scale, and DQ marks an arm a disqualification gate refused a rank (main paper Tab. 4). No generated build shown is a memorized copy: none of the 48 generated cells reaches IoU 0.95 to a training build. The contaminated checkpoint of Fig. 11 is absent: it is unrankable, and a gallery ordered by score would imply a standing

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